Hyperspectral Image Denoising Using Spectral-Spatial Transform-Based Sparse and Low-Rank Representations

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***Abstract*— This article proposes a denoising method based on sparse spectral–spatial and low-rank representations (SSSLRR) using the 3-D orthogonal transform (3-DOT). SSSLRR can be effectively used to remove the Gaussian and mixed noise. SSSLRR uses 3-DOT to decompose noisy HSI to sparse transform coefficients. The 3-D discrete orthogonal wavelet transform (3-D DWT) is a representative 3-DOT suitable for denoising since it concentrates on the signal in few transform coefficients, and the 3-D discrete orthogonal cosine transform (3-D DCT) is another example. An SSSLRR using 3-D DWT will be called SSSLRR-DWT. SSSLRR-DWT is an iterative algorithm based on the alternating direction method of multipliers (ADMM) that uses sparse and nuclear norm penalties. We use an ablation study to show the effectiveness of the penalties we employ in the method. Both simulated and real hyperspectral datasets demonstrate that SSSLRR outperforms other comparative methods in quantitative and visual assessments to remove the Gaussian and mixed noise.**

***Index Terms*— Denoising, hyperspectral image, low rank, orthogonal transform, sparse.**

I. Introduction

HYPERSPECTRAL images (HSIs) acquired by hyperspectral imaging sensors contain hundreds of spectral bands. The spectral information an HSI provides makes it possible to discriminate different materials in a scene. Therefore, HSIs have been used in applications, such as agricultural production, mineral exploration, and environmental protection [1]–[4]. However, due to photon effects, atmospheric absorption, and sensor disturbance, HSIs are corrupted by different kinds of noises, such as quantization noise, thermal noise, and shot noise. Therefore, denoising is an essential preprocessing step for HSI analysis and applications.

HSI denoising has evolved considerably in recent years [5]–[12]. Grayscale image 2-D denoising methods can be extended to HSI denoising by treating each spectral band separately. Examples are the nonlocal means (NLM) filter [13], the 2-D wavelet denoising [14], the block-matching and 3-D filter (BM3D) [15], and the weighted nuclear norm minimization (WNNM) [16]. Single-band methods utilize only spatial information leading to spectral distortion.

HSIs are spectrally correlated. Therefore, denoising methods should jointly exploit spatial and spectral information [17]. Several spectral–spatial denoising methods have been proposed [18]. Yuan *et al.* [19] suggested a spectral–spatial adaptive total variation denoising method for removing the Gaussian noise. Chen *et al.* [20] proposed a denoising method based on spectral–spatial domain mixing prior. Huang *et al.* [21] used a spatial–spectral weighted nuclear minimization denoising method for removing mixed types of noise in HSI. Besides all the methods above, there are other methods based on transforming HSI to the wavelet domain and utilize the wavelet-based spatial and spectral information to remove the noise [22]. Chen and Qian [23] used a principal component analysis (PCA) of the HSI and then transformed the low-energy PCA output channels to wavelet domain to remove the noise using 2-D spatial and 1-D spectral wavelet shrinkage denoising. Othman and Qian [24] suggested a hybrid of spatial and spectral wavelet shrinkage by using a spectral derivative to elevate noise and then utilized wavelet-based spatial and spectral denoising. Rasti *et al.* [25] presented a wavelet domain-based spectral–spatial penalty method for HSI denoising. These wavelet-based spectral–spatial methods use the spatial and spectral information of HSI to remove noise and utilize wavelet transform to provide a sparse representation for clean HSI to further remove noise in the wavelet domain.

Recently, many low-rank model-based methods have been proposed for HSI denoising [26]–[32]. Zhang *et al.* [33] presented a low-rank matrix recovery (LRMR) method for removing mixed types of noise in HSI. An extension of the LRMR method [34] was proposed using an adaptive iteration regularization framework for HSI denoising. Chen *et al.* [35] presented a nonconvex low-rank matrix approximation method using a nonconvex regularizer. These low-rank model-based methods exploit the low-dimensional structures in the HSIs. HSIs are usually self-similarity and have sparse characteristics in the spatial domain and hence can be denoised using sparse

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representation. Lu *et al.* [36] and Li *et al.* [37] suggested an HSI denoising method based on sparse representation using the high similar spatial information of HSI. More recently, several denoising methods based on a combination of low-rank and sparse representation have been presented. Liu *et al.* [38] used group sparse and nuclear norm low-rank penalties for removing stripe noise. Zhuang and Bioucas-Dias [39] utilized similar patches in the low-dimensional subspace to exploit the sparse and low-rank characteristics of HSI. Rasti *et al.* [40] used a sparse low-rank model, which utilizes sparse penalized least squares for estimating the unknown signal for denoising. Tensor-based denoising methods given in [41]–[49] exploited the low rank and the sparse characteristic of similar tensors to remove noise.

In this article, we propose a denoising method based on sparse spectral–spatial and low-rank representations (SSSLRR) using the 3-D orthogonal transform (3-DOT) for HSI denoising to remove the Gaussian and mixed noise. This article used both 3-D discrete orthogonal wavelet transform (3-D DWT) and 3-D discrete orthogonal cosine transform (3-D DCT) as a representative of 3-DOT. The 3-D DWT is exploited in SSSLRR-DWT to provide a sparse representation for HSI by jointly using the correlated spatial and spectral information. The idea is that the wavelet transform concentrates the signal energy in few coefficients, while the noise has energy spread over all coefficients [50], [51]. A new sparse and low-rank penalized model is proposed for removing noise in the wavelet domain. The method uses the ℓ_1 penalty and the weighted nuclear norm low-rank penalty to adaptively shrink the wavelet coefficients and penalize the singular values to remove the noise. The main contributions of this article are summarized as follows.

1) A denoising method based on SSSLRR using 3-D orthogonal transform (3-DOT) is proposed for HSI denoising and is effectively used to remove the Gaussian and mixed noise. Examples of 3-D DOT used are 3-D DWT and 3-D DCT.
2) A new sparse and low-rank penalized model is proposed in our SSSLRR. Unlike the traditional spectral–spatial domain sparse and low-rank model, our new sparse and low-rank penalized model is solved using an iterative algorithm for a global image and uses both ℓ_1 and weighted nuclear norm low-rank penalties applied in the 3-D DOT domain.

An algorithm based on the alternating direction method of multipliers (ADMM) is developed to get the optimal parameters and denoised results for SSSLRR.

The effectiveness of different penalties for SSSLRR-DWT is analyzed using four different methods both in signal and wavelet domains. The main conclusion is that SSSLRR-DWT using both ℓ_1 and weighted nuclear norm low-rank penalties in the wavelet domain yields better denoising results than the other methods used in experiments. Moreover, an SSSLRR-DCT denoising method where the 3-D DCT is used instead of using 3-D DWT is also proposed and analyzed. The results show that both SSSLRR-DWT and SSSLRR-DCT yield good denoised performance for removing Gaussian and mixed noise.

The rest of this article is organized as follows. Section II formulates and solves the SSSLRR HSI denoising problem using ADMM and analyzes the effectiveness of SSSLRR-DWT. Section III presents and analyzes extensive experimental results. Finally, Section IV concludes this article.

II. Problem Formulation and Proposed Denoiser

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_B] \in \mathbb{R}^{n\times B}$ denote a clean HSI with n pixels and B spectral bands. Here, $\mathbf{x}_i \in \mathbb{R}^n$ is the ith vectorized spectral band. We assume the following observational model:

$$\mathbf{Y} = \mathbf{X} + \mathbf{N} \tag{1}$$

where $\mathbf{Y}$, $\mathbf{N} \in \mathbb{R}^{n\times B}$ represent the observed noisy HSI and noise, respectively. Therefore, estimating $\widehat{\mathbf{X}} \in \mathbb{R}^{n\times B}$ of $\mathbf{X}$ from $\mathbf{Y}$ is the HSI denoising problem.

A. SSSLRR

Using the 3-DOT decomposition, the observational model is formulated as

$$\mathbf{Y} = \mathbf{D}_2\mathbf{W}\mathbf{D}_1^T + \mathbf{N} \tag{2}$$

where $\mathbf{D}_2 \in \mathbb{R}^{n\times n}$ represents a spatial part of the 3-DOT, $\mathbf{D}_1 \in \mathbb{R}^{B\times B}$ represents a spectral part of the 3-DOT, and $\mathbf{W} \in \mathbb{R}^{n\times B}$ is a coefficients matrix.

To enforce sparsity and low-rank, a cost function using an ℓ_1 and a weighted nuclear norm low-rank penalties is defined as

$$\widehat{\mathbf{W}} = \arg\min_{\mathbf{W}} \ \|\mathbf{H} - \mathbf{W}\|_F^2 + \lambda_1\|\mathbf{W}\|_{1,1} + \|\mathbf{W}\|_{\omega,*} \tag{3}$$

where $\mathbf{H} = \mathbf{D}_2^T\mathbf{Y}\mathbf{D}_1$, $\|\cdot\|_F^2$ is the Frobenius norm, $\|\mathbf{W}\|_{1,1} := \sum_{i=1}^{B}\|\mathbf{w}_i\|_1$ ($\mathbf{w}_i$ denotes the ith column of $\mathbf{W}$), and $\|\mathbf{W}\|_{\omega,*}$ is the weighted nuclear norm of $\mathbf{W}$, which is defined as

$$\|\mathbf{W}\|_{\omega,*} = \Sigma_i|\omega_i\sigma_i(\mathbf{W})| \tag{4}$$

where ω_i is the weight assigned to the ith singular value $\sigma_i(\mathbf{W})$ of $\mathbf{W}$.

We adapt the alternating direction method of multipliers (ADMM) [52] for solving (3). The optimization problem (3) is transformed into the equivalent problem

$$\begin{aligned}&\min_{\mathbf{W},\mathbf{Z}} \ \|\mathbf{H} - \mathbf{W}\|_F^2 + \lambda_1\|\mathbf{W}\|_{1,1} + \|\mathbf{Z}\|_{\omega,*}\\ &\text{s.t. } \mathbf{W} - \mathbf{Z} = \mathbf{0}.\end{aligned} \tag{5}$$

The augmented Lagrangian function of the abovementioned optimization is

$$\begin{aligned}L_\mu(\mathbf{W},\mathbf{Z},\mathbf{U}) = \|\mathbf{H} - \mathbf{W}\|_F^2 + \lambda_1\|\mathbf{W}\|_{1,1} + \|\mathbf{Z}\|_{\omega,*}\\ +\frac{\mu}{2}\|\mathbf{W} - \mathbf{Z} + \mathbf{U}\|_F^2\end{aligned} \tag{6}$$

where $\mu > 0$ is the ADMM penalty parameter. The ADMM consists of the iterations.

W-Step: Fix $\mathbf{Z}^k$, $\mathbf{U}^k$ and estimate $\mathbf{W}^{k+1}$ by solving

$$\mathbf{W}^{k+1} := \arg\min_{\mathbf{W}} \ L_\mu(\mathbf{W},\mathbf{Z}^k,\mathbf{U}^k). \tag{7}$$

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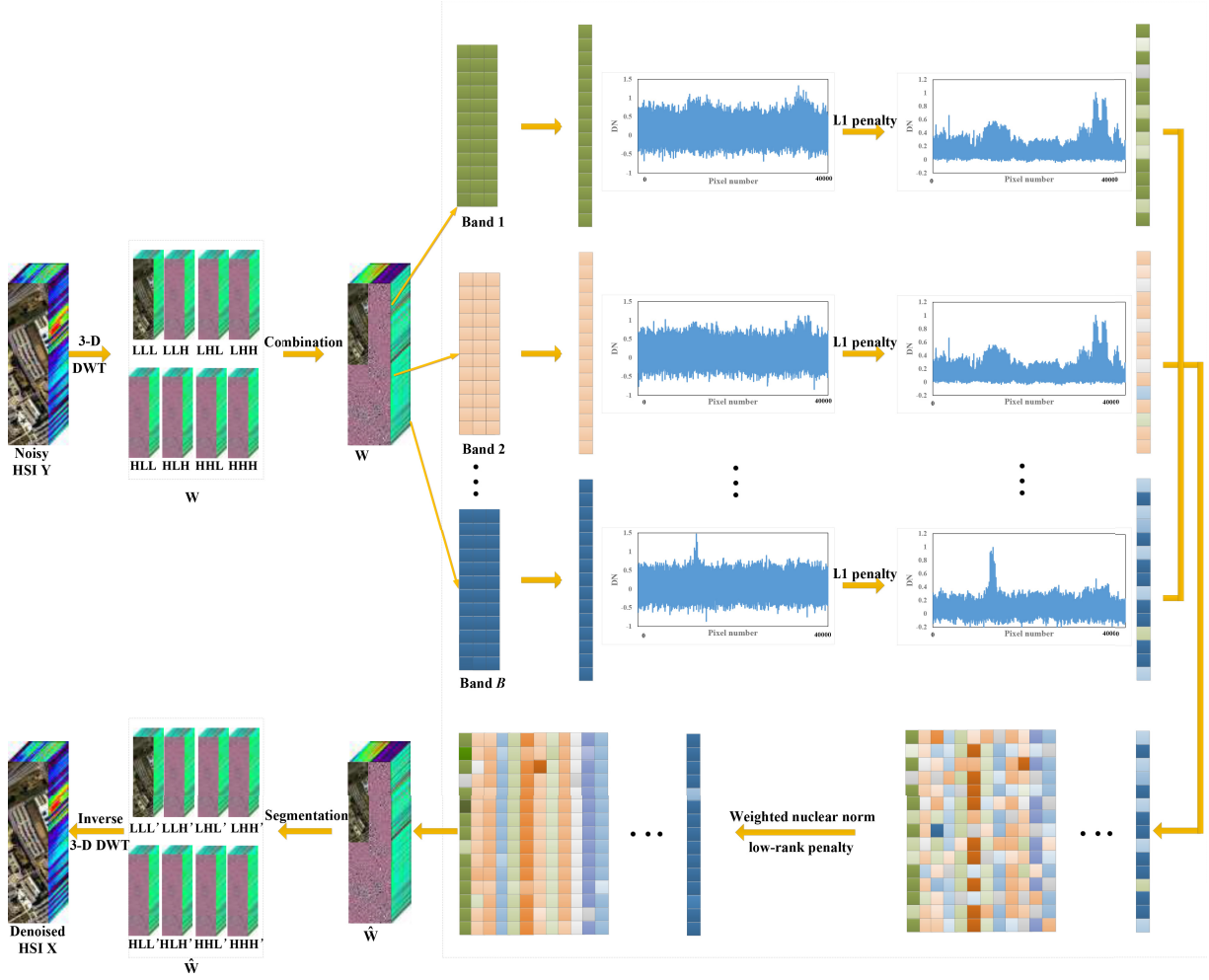


Fig. 1. Schematic of the SSSLRR, which uses the 3-D DWT for the 3-DOT.

Z-Step: Fix $\mathbf{W}^{k+1}$, $\mathbf{U}^k$ and estimate $\mathbf{Z}^{k+1}$ by solving

$$\mathbf{Z}^{k+1} := \arg\min_{\mathbf{Z}} \; L_\mu(\mathbf{W}^{k+1}, \mathbf{Z}, \mathbf{U}^k). \tag{8}$$

U-Step: Fix $\mathbf{W}^{k+1}$, $\mathbf{Z}^{k+1}$ and update $\mathbf{U}^{k+1}$ with

$$\mathbf{U}^{k+1} := \mathbf{U}^k + \mathbf{W}^{k+1} - \mathbf{Z}^{k+1}. \tag{9}$$

The minimization problem (7) is equivalent to

$$\arg\min_{\mathbf{W}} \; \|\mathbf{H} - \mathbf{W}\|_F^2 + \lambda_1 \|\mathbf{W}\|_{1,1} + \frac{\mu}{2}\|\mathbf{W} - \mathbf{Z}^k + \mathbf{U}^k\|_F^2. \tag{10}$$

By ignoring irrelevant terms and using trace properties, the minimization problem (10) is equivalent to

$$\arg\min_{\mathbf{W}} \; \frac{1}{2}\|\mathbf{W} - \mathbf{B}\|_F^2 + \lambda \|\mathbf{W}\|_{1,1} \tag{11}$$

where $\mathbf{B} = (2\mathbf{H} + \mu\mathbf{Z}^k - \mu\mathbf{U}^k)/(2+\mu)$ and $\lambda = \lambda_1/(2+\mu)$. This problem (11) can be solved by using the soft-thresholding function [53]

$$w_{i,j}^{k+1} = \text{sign}(b_{i,j}) \max\{|b_{i,j}| - \lambda, 0\} \tag{12}$$

where $w_{i,j}^{k+1}$ and $b_{i,j}$ are the entries in the ith row and jth column of $\mathbf{W}^{k+1}$ and $\mathbf{B}$, respectively.

Algorithm 1 SSSLRR HSI Denoising Algorithm

Input: $\mathbf{Y}$, λ_1, λ_ω, and μ
Initialization: $\mathbf{Z}^0 = \mathbf{U}^0 = 0$, and $k = 0$
repeat
  $\mathbf{B} = \frac{2\mathbf{D}_2^T\mathbf{Y}\mathbf{D}_1 + \mu\mathbf{Z}^k - \mu\mathbf{U}^k}{2+\mu}$
  $\mathbf{W}^{k+1} = \arg\min_{\mathbf{W}} \; \frac{1}{2}\|\mathbf{W} - \mathbf{B}\|_F^2 + \frac{\lambda_1}{2+\mu}\|\mathbf{W}\|_{1,1}$
  $\mathbf{Z}^{k+1} = \arg\min_{\mathbf{Z}} \; \frac{\mu}{2}\|\mathbf{W}^{k+1} - \mathbf{Z} + \mathbf{U}^k\|_F^2 + \|\mathbf{Z}\|_{\omega,*}$
  $\mathbf{U}^{k+1} = \mathbf{U}^k + \mathbf{W}^{k+1} - \mathbf{Z}^{k+1}$
  $k \leftarrow k + 1$
until stopping criterion is satisfied, i.e., $\|\mathbf{W}^{k+1} - \mathbf{W}^k\|_F^2 / \|\mathbf{W}^k\|_F^2 \le \epsilon$
Output: $\widehat{\mathbf{X}} = \mathbf{D}_2\widehat{\mathbf{W}}\mathbf{D}_1^T$

The minimization problem (8) is equivalent to

$$\mathbf{Z}^{k+1} = \arg\min_{\mathbf{Z}} \; \frac{\mu}{2}\|\mathbf{W}^{k+1} - \mathbf{Z} + \mathbf{U}^k\|_F^2 + \|\mathbf{Z}\|_{\omega,*} \tag{13}$$

has the closed-form solution [16]

$$\mathbf{Z}^{k+1} = \mathbf{P}\mathbf{S}_\omega(\Sigma)\mathbf{Q}^T \tag{14}$$

where $\mathbf{W}^{k+1} + \mathbf{U}^k = \mathbf{P}\Sigma\mathbf{Q}^T$ is the singular value decomposition (SVD) of $\mathbf{W}^{k+1} + \mathbf{U}^k$ with singular values ordered

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TABLE I

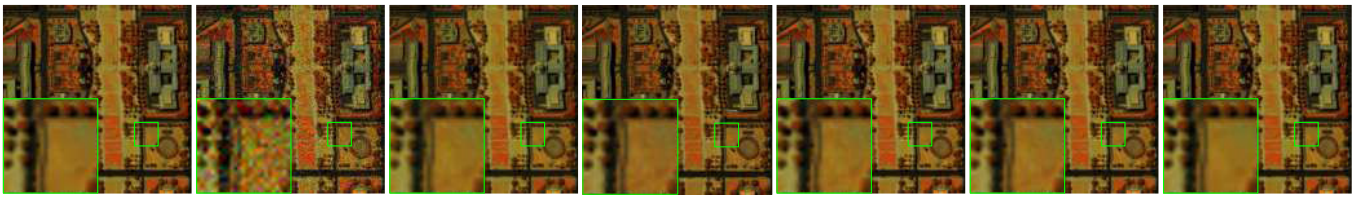
QUANTITATIVE EVALUATION RESULTS [PSNR (dB), MSSIM, MFSIM, AND SAM (DEGREES)] FOR THE DIFFERENT DENOISING METHODS USING THE WASHINGTON DC MALL (WDCM) DATASET. THE BEST RESULTS ARE IN BOLD TYPEFACE

Noise level	Metric	Noisy	SSWSR	SSWLRR1	SSWLRR2	SLRR	SSSLRR-DWT
$\sigma = 0.1$	PSNR	20.004	32.171	33.304	35.627	35.906	**37.968**
	MSSIM	0.4528	0.9263	0.9449	0.9582	0.9593	**0.9794**
	MFSIM	0.7348	0.9598	0.9707	0.9745	0.9790	**0.9867**
	SAM	30.350	8.5714	7.4659	6.0733	6.0960	**4.0887**
$\sigma = 1$, $\eta = 10$	PSNR	22.820	32.429	31.771	33.965	38.267	**44.165**
	MSSIM	0.8674	0.9633	0.9553	0.9691	0.9857	**0.9968**
	MFSIM	0.9306	0.9777	0.9701	0.9781	0.9910	**0.9977**
	SAM	23.863	8.9575	7.7902	6.0998	4.8411	**1.9880**
$\sigma \sim U(0.1, 0.2)$	PSNR	16.371	28.816	30.774	32.777	33.250	**35.544**
	MSSIM	0.3083	0.8686	0.9199	0.9274	0.9326	**0.9661**
	MFSIM	0.6423	0.9289	0.9565	0.9573	0.9655	**0.9782**
	SAM	39.689	12.054	9.1468	7.9920	8.0173	**5.2188**

TABLE II

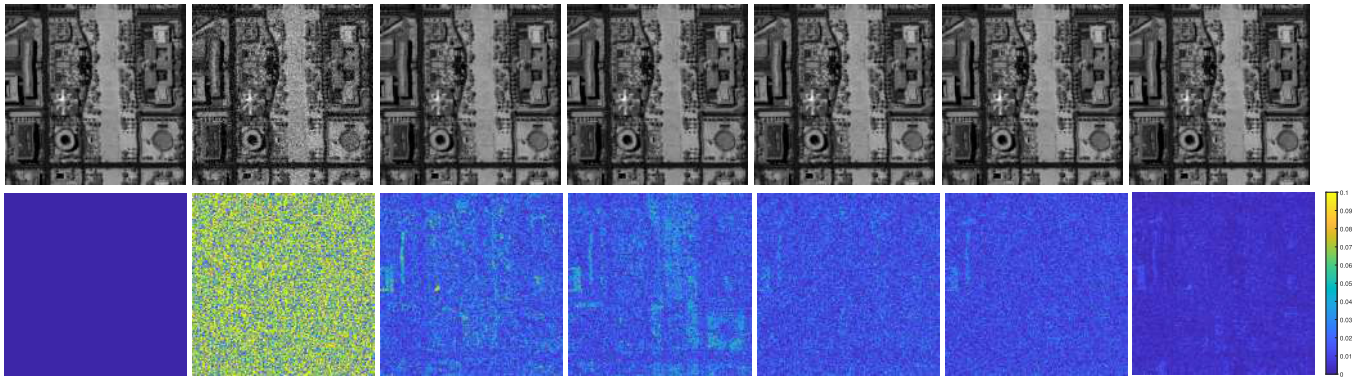
QUANTITATIVE EVALUATION RESULTS [PSNR (dB), MSSIM, MFSIM, AND SAM (DEGREES)] FOR THE DIFFERENT DENOISING METHODS USING THE UNIVERSITY OF PAVIA (UP) DATASET. THE BEST RESULTS ARE IN BOLD TYPEFACE

Noise level	Metric	Noisy	SSWSR	SSWLRR1	SSWLRR2	SLRR	SSSLRR-DWT
Case 7	PSNR	12.891	23.738	24.054	26.119	28.432	**30.933**
	MSSIM	0.1465	0.6590	0.6876	0.7489	0.7883	**0.9066**
	MFSIM	0.5321	0.8601	0.8735	0.8902	0.9169	**0.9553**
	SAM	41.441	12.021	12.124	11.309	8.8532	**7.2942**



(a) (b) (c) (d) (e) (f) (g)

Fig. 2. Denoising for the WDCM dataset, Case 1 with $\sigma = 0.1$, using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 70, G: 120, and B: 170), (b) noisy image, (c) SSWSR, (d) SSWLRR1, (e) SSWLRR2, (f) SLRR, and (g) SSSLRR-DWT.



(a) (b) (c) (d) (e) (f) (g)

Fig. 3. Denoised images (the first row) and residual images (the second row) for the WDCM dataset, Case 1 with $\sigma = 0.1$, using different methods. (a) Original band 95, (b) noisy band 95, (c) SSWSR, (d) SSWLRR1, (e) SSWLRR2, (f) SLRR, and (g) SSSLRR-DWT.

by nonincreasing magnitude. $\mathbf{S}_\omega(\Sigma)$ is a generalized soft-thresholding function on the diagonal matrix Σ. For the ith singular value Σ_{ii}

$$\mathbf{S}_\omega(\Sigma)_{ii} = \max(\Sigma_{ii} - \omega_i, 0). \tag{15}$$

The weight ω_i is

$$\omega_i = \frac{\lambda_\omega}{\mu(\Sigma_{ii} + \varepsilon)} \tag{16}$$

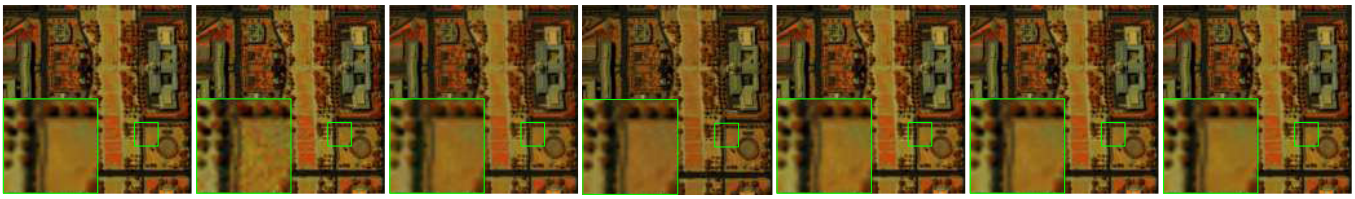
where ε is a nonzero constant and is used to avoid dividing by zero and λ_ω is a constant. This means that larger singular

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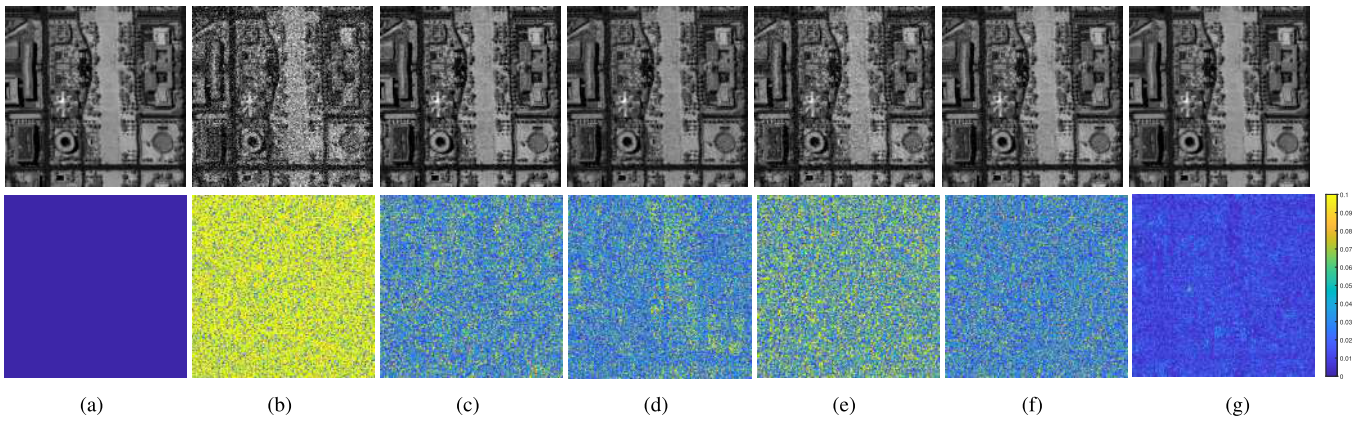
(a) (b) (c) (d) (e) (f) (g)

Fig. 4. Denoising for the UP dataset, Case 7 using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 64, G: 45, and B: 10), (b) noisy image, (c) SSWSR, (d) SSWLRR1, (e) SSWLRR2, (f) SLRR, and (g) SSSLRR-DWT.



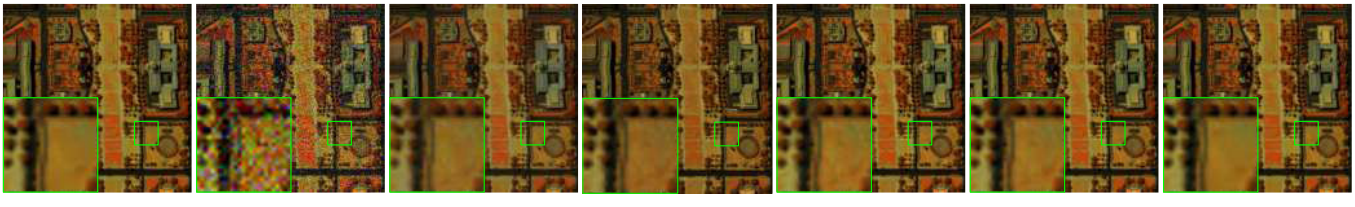
(a) (b) (c) (d) (e) (f) (g)

Fig. 5. Denoising for the WDCM dataset, Case 2 with $\sigma = 1$, $\eta = 10$, using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 70, G: 120, B: 170), (b) noisy image, (c) SSWSR, (d) SSWLRR1, (e) SSWLRR2, (f) SLRR, and (g) SSSLRR-DWT.



(a) (b) (c) (d) (e) (f) (g)

Fig. 6. Denoised images (the first row) and residual images (the second row) for the WDCM dataset, Case 2 with $\sigma = 1$, $\eta = 10$, using different methods. (a) Original band 95, (b) noisy band 95, (c) SSWSR, (d) SSWLRR1, (e) SSWLRR2, (f) SLRR, and (g) SSSLRR-DWT.



(a) (b) (c) (d) (e) (f) (g)

Fig. 7. Denoising for the WDCM dataset, Case 3 with $\sigma \sim U(0.1, 0.2)$, using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 70, G: 120, and B: 170), (b) noisy image, (c) SSWSR, (d) SSWLRR1, (e) SSWLRR2, (f) SLRR, and (g) SSSLRR-DWT.

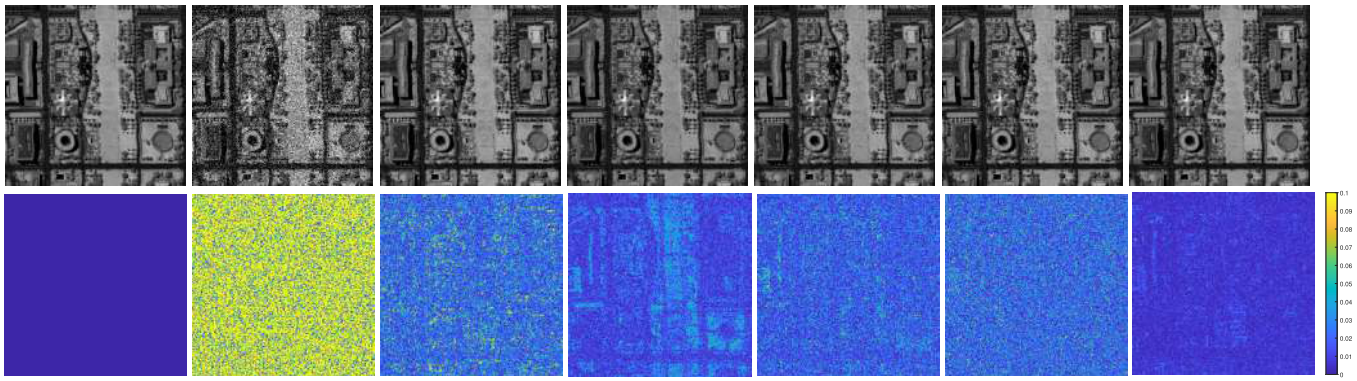
values will be penalized less than smaller singular values. Thus, the weighted nuclear norm low-rank penalty provides better estimation for the rank compared with the nuclear norm, which treats all the singular values equally, since larger singular values implying major information that are more important than smaller ones [16].

Finally, the estimate is obtained as

$$\widehat{\mathbf{X}} = \mathbf{D}_2 \widehat{\mathbf{W}} \mathbf{D}_1^T. \tag{17}$$

The SSSLRR method is summarized in Algorithm 1 and shown in Fig. 1.

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(a) (b) (c) (d) (e) (f) (g)

Fig. 8. Denoised images (the first row) and residual images (the second row) for the WDCM dataset, Case 3 with $\sigma \sim U(0.1, 0.2)$, using different methods. (a) Original band 95, (b) noisy band 95, (c) SSWSR, (d) SSWLRR1, (e) SSWLRR2, (f) SLRR, and (g) SSSLRR-DWT.

TABLE III

QUANTITATIVE EVALUATION RESULTS [PSNR (dB), MSSIM, MFSIM, SAM (DEGREES), AND TIME (SECOND)] FOR THE DIFFERENT DENOISING METHODS USING THE WASHINGTON DC MALL (WDCM) DATASET. THE BEST RESULTS ARE IN BOLD TYPEFACE

WDCM	Noise level	Metric	Noisy	BM3D	BM4D	LRTA	NAIRLMA	TDL	KBR	HyRes	FastHyDe	SSSLRR-DWT
Case 1	$\sigma = 0.05$	PSNR	26.025	31.031	36.624	37.035	40.288	39.615	38.617	40.297	41.219	**41.681**
		MSSIM	0.7191	0.8948	0.9677	0.9699	0.9862	0.9855	0.9830	0.9870	0.9898	**0.9908**
		MFSIM	0.8691	0.9365	0.9808	0.9829	0.9919	0.9910	0.9903	0.9921	0.9935	**0.9942**
		SAM	18.048	8.6547	5.1559	5.1145	3.6577	3.6349	3.0349	3.4451	2.9787	**2.8475**
		Time	-	118.12	462.12	1.5192	58.323	8.0064	1253.3	1.5588	**0.3309**	11.754
	$\sigma = 0.1$	PSNR	20.004	27.429	32.468	33.181	36.109	35.581	34.582	36.419	37.671	**37.968**
		MSSIM	0.4528	0.7778	0.9209	0.9370	0.9661	0.9657	0.9596	0.9701	0.9779	**0.9794**
		MFSIM	0.7348	0.8675	0.9541	0.9612	0.9806	0.9792	0.9758	0.9821	0.9858	**0.9867**
		SAM	30.350	10.742	7.5690	6.6810	5.6988	5.3089	4.4250	5.2171	4.1810	**4.0887**
		Time	-	137.68	461.66	1.4653	48.741	7.8660	1337.1	1.2664	**0.3454**	12.593
	$\sigma = 0.2$	PSNR	13.983	24.199	28.540	29.300	31.653	31.280	30.145	32.442	33.948	**34.518**
		MSSIM	0.2061	0.5929	0.8197	0.8675	0.9177	0.9178	0.8956	0.9324	0.9528	**0.9582**
		MFSIM	0.5644	0.7482	0.8994	0.9215	0.9548	0.9524	0.9383	0.9599	0.9700	**0.9734**
		SAM	46.309	12.615	10.651	8.3784	8.4539	7.1401	6.2672	7.0750	5.9433	**5.6292**
		Time	-	120.97	463.11	1.7205	42.016	7.5671	2644.2	1.3871	**0.3403**	13.466
Case 2	$\sigma = 0.5$, $\eta = 10$	PSNR	28.840	34.056	39.026	29.008	43.879	29.006	35.239	46.498	42.381	**46.750**
		MSSIM	0.9321	0.9707	0.9890	0.9330	0.9955	0.9336	0.9713	0.9978	0.9935	**0.9979**
		MFSIM	0.9641	0.9808	0.9926	0.9643	0.9969	0.9649	0.9832	0.9984	0.9955	**0.9985**
		SAM	13.704	6.7747	3.9992	13.476	2.3161	13.471	6.2553	1.5969	2.5904	**1.5664**
		Time	-	118.91	464.55	1.3040	100.21	9.8575	1223.6	1.6714	**0.2571**	44.097
	$\sigma = 1$, $\eta = 10$	PSNR	22.820	30.645	35.060	23.096	40.088	22.987	29.944	43.602	39.003	**44.165**
		MSSIM	0.8674	0.9401	0.9754	0.8680	0.9912	0.8694	0.9329	0.9965	0.9864	**0.9968**
		MFSIM	0.9306	0.9636	0.9845	0.9307	0.9937	0.9316	0.9634	0.9975	0.9906	**0.9977**
		SAM	23.863	8.9355	5.9647	23.266	3.3176	23.530	11.356	2.1116	3.5566	**1.9880**
		Time	-	116.93	463.54	2.0748	100.27	10.306	1233.6	1.5828	**0.3015**	44.075
	$\sigma = 1$, $\eta = 20$	PSNR	22.815	29.410	34.324	24.515	38.917	24.686	34.515	39.877	38.928	**40.436**
		MSSIM	0.7853	0.9087	0.9668	0.8188	0.9873	0.8314	0.9629	0.9902	0.9859	**0.9913**
		MFSIM	0.8927	0.9454	0.9790	0.9034	0.9915	0.9148	0.9773	0.9931	0.9903	**0.9939**
		SAM	23.972	9.4715	6.2886	20.424	3.7875	20.236	6.0661	3.1943	3.5651	**3.0618**
		Time	-	115.41	465.22	4.6855	99.264	9.5967	1233.0	1.4268	**0.3048**	36.186
Case 3	$\sigma \sim U$ (0.1,0.2)	PSNR	16.371	25.161	30.069	23.942	33.646	32.977	31.668	34.509	35.331	**35.544**
		MSSIM	0.3083	0.6624	0.8694	0.6447	0.9455	0.9384	0.9256	0.9546	0.9645	**0.9661**
		MFSIM	0.6423	0.8008	0.9258	0.8075	0.9691	0.9651	0.9562	0.9732	0.9773	**0.9782**
		SAM	39.689	12.821	9.3562	20.305	7.0918	6.8392	5.5020	6.1270	5.3139	**5.2188**
		Time	-	119.62	461.44	4.5692	40.548	8.1913	2276.4	1.3455	**0.3276**	12.620

B. SSSLRR-DWT

The SSSLRR-DWT denoising method uses the 3-D discrete orthogonal wavelet transform (3-D DWT) to be a representative of the 3-DOT. In SSSLRR-DWT, the 2-D orthogonal (spatial) wavelet basis $\mathbf{D}_2 \in \mathbb{R}^{n\times n}$ and the 1-D orthogonal (spectral) wavelet basis $\mathbf{D}_1 \in \mathbb{R}^{B\times B}$ are used for 3-D DWT to decompose the 3-D HSI to sparse wavelet coefficients $\mathbf{W} \in \mathbb{R}^{n\times B}$. SSSLRR-DWT uses the sparse and low-rank penalized model for the wavelet coefficients to remove noise.

C. SSSLRR-DCT

The 3-D discrete orthogonal cosine transform (3-D DCT) is also a classical 3-DOT. The SSSLRR-DCT denoising method uses 3-D DCT as the 3-DOT in SSSLRR. For SSSLRR-DCT,

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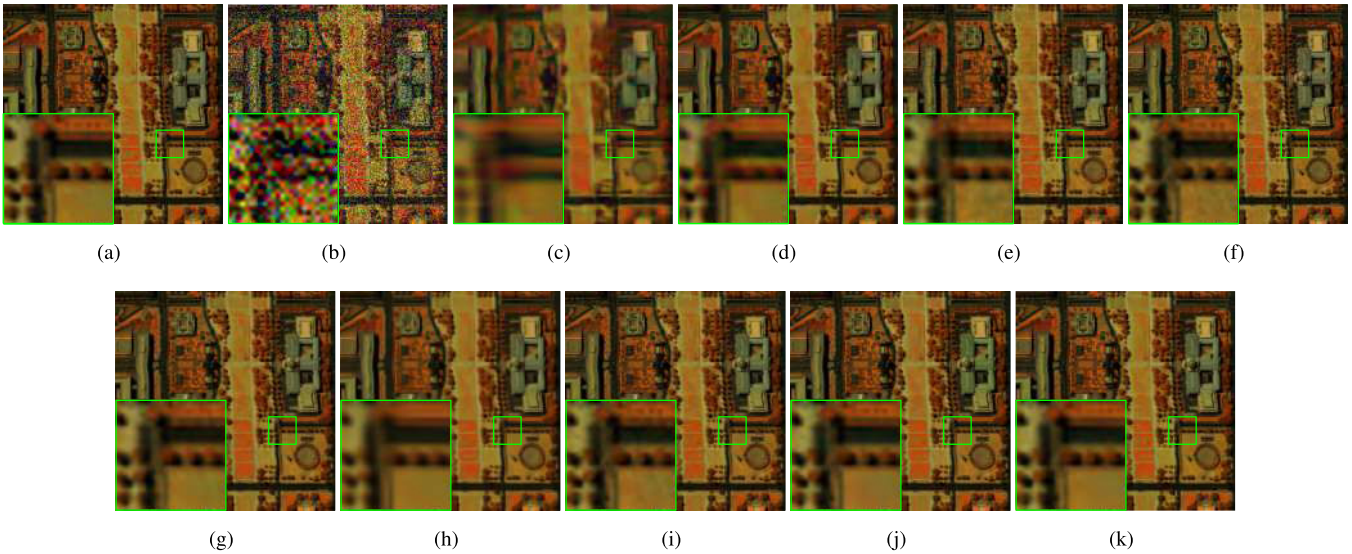


Fig. 9. Denoising for the WDCM dataset, Case 1 with $\sigma = 0.2$, using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 70, G: 120, and B: 170), (b) noisy image, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

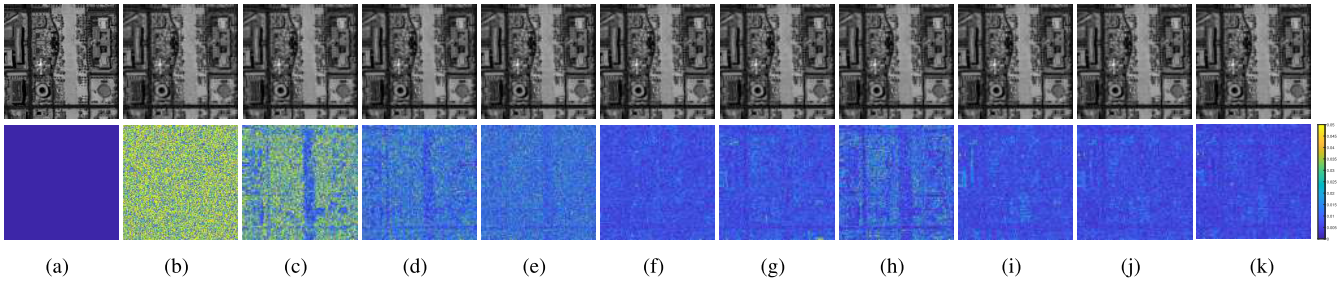


Fig. 10. Denoised images (the first row) and residual images (the second row) for the WDCM dataset, Case 1 with $\sigma = 0.05$, using different methods. (a) Original band 95, (b) noisy band 95, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

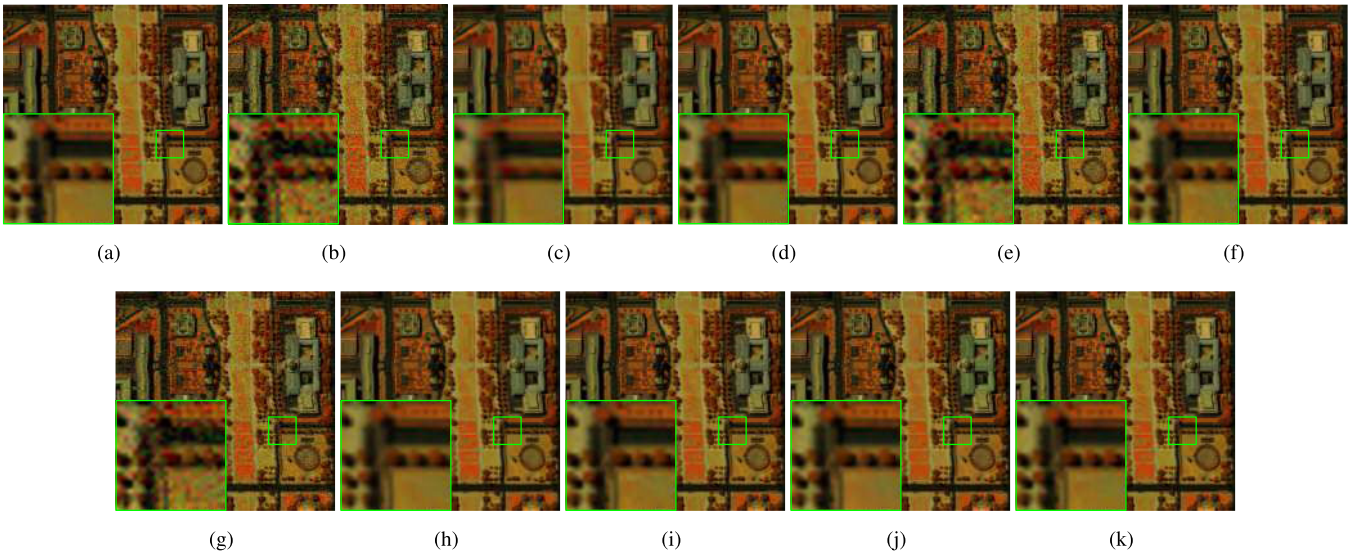


Fig. 11. Denoising for the WDCM dataset, Case 2 with $\sigma = 1$, $\eta = 20$ using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 70, G: 120, and B: 170), (b) noisy image, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

the 2-D orthogonal (spatial) cosine basis $\mathbf{D}_2 \in \mathbb{R}^{n\times n}$ and the 1-D orthogonal (spectral) cosine basis $\mathbf{D}_1 \in \mathbb{R}^{B\times B}$ are used for 3-D DCT to decompose the noisy HSI to sparse cosine coefficients $\mathbf{W} \in \mathbb{R}^{n\times B}$. The sparse and low-rank penalized model is utilized for SSSLRR-DCT to shrink the cosine coefficients to remove noise.

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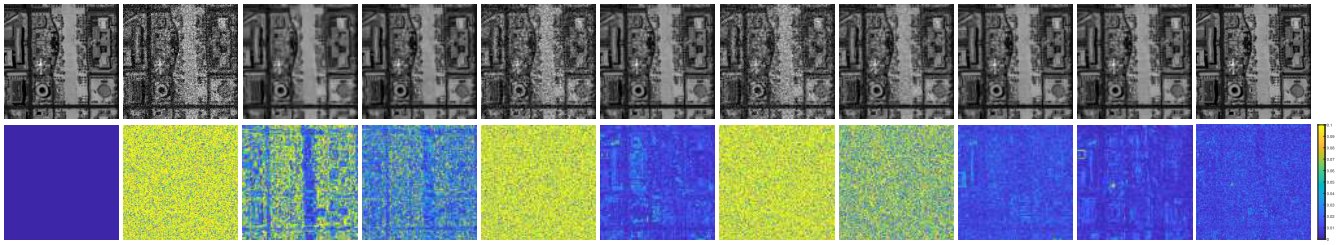


Fig. 12. Denoised images (the first row) and residual images (the second row) for the WDCM dataset, Case 2 with $\sigma = 1$, $\eta = 10$, using different methods. (a) Original band 95, (b) noisy band 95, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

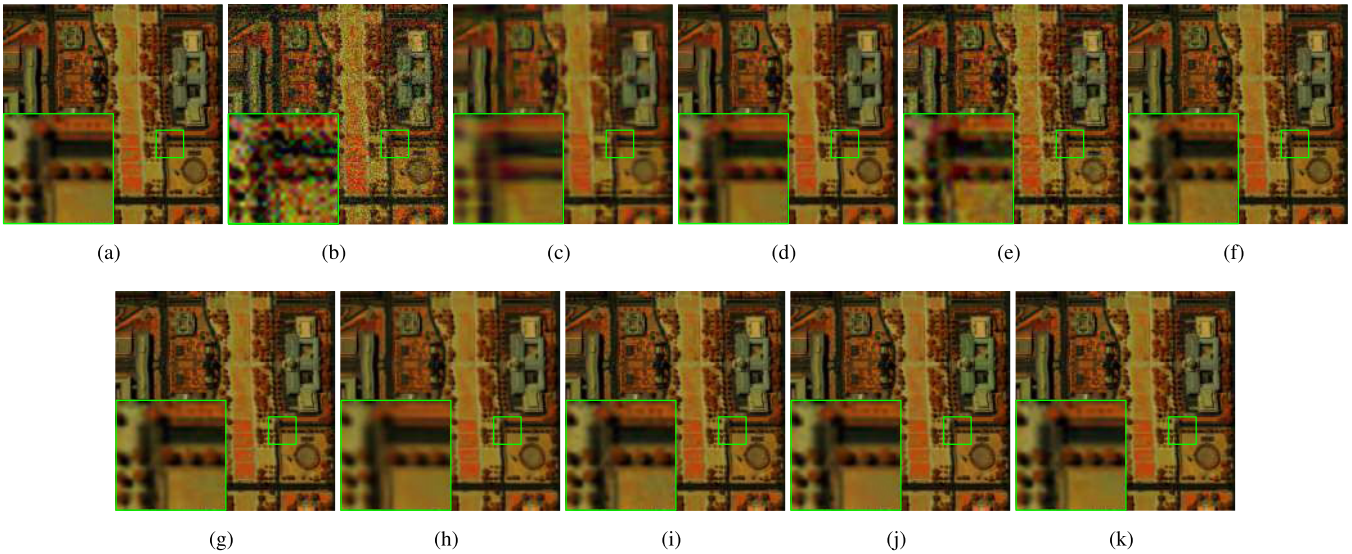


Fig. 13. Denoising for the WDCM dataset, Case 3 with $\sigma \sim U(0.1, 0.2)$, using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 70, G: 120, and B: 170), (b) noisy image, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

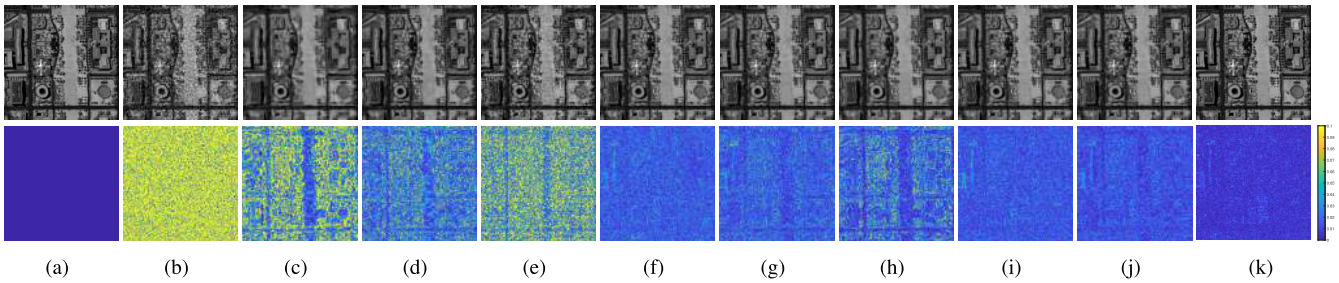


Fig. 14. Denoised images (the first row) and residual images (the second row) for the WDCM dataset, Case 3 with $\sigma \sim U(0.1, 0.2)$, using different methods. (a) Original band 95, (b) noisy band 95, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

The 3-D DWT or 3-D DCT can concentrate the signal energy in few coefficients, while the noise spreads over all coefficients. Both SSSLRR-DWT and SSSLRR-DCT are iterative algorithms and can adaptively shrink either wavelet coefficients or cosine transform coefficients and penalize the singular values of the coefficient matrix. The larger (signal) singular values will be shrunk less, while the smaller ones (noise) will be shrunk more leading to denoising. Both SSSLRR-DWT and SSSLRR-3DCT yield good denoising performance. However, the focus of this article is on SSSLRR-DWT.

D. Ablation Study

SSSLRR-DWT uses the sparse and low-rank penalized model based on the ℓ_1 and weighted nuclear norm low-rank penalties in the wavelet domain for HSI denoising. To further analyze the effectiveness of the penalties for the

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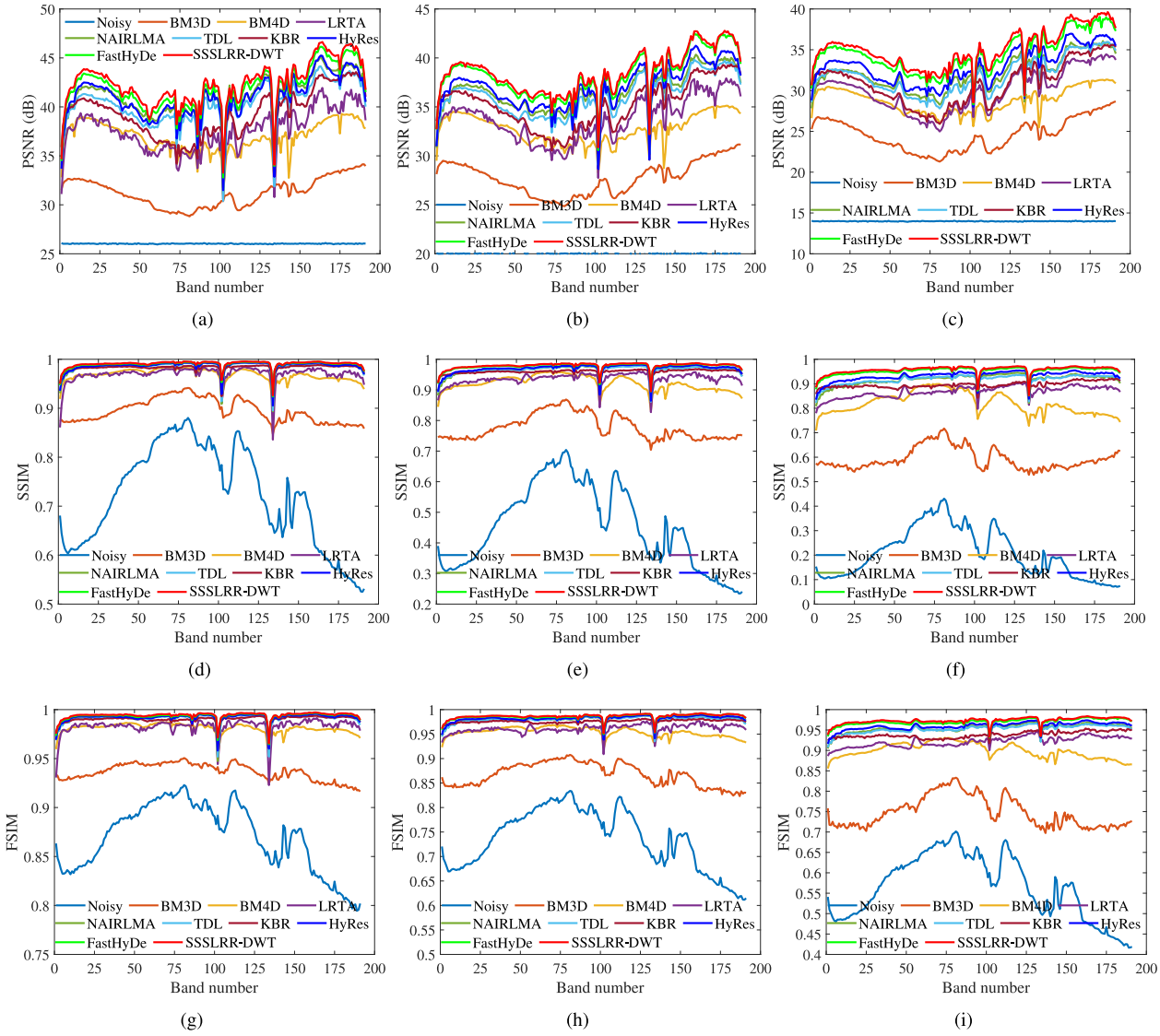


Fig. 15. (a)–(c) PSNR, (d)–(f) SSIM, and (g)–(i) FSIM values of each band of the denoised WDCM dataset. The first-to-third columns correspond to Case 1 with $\sigma = 0.05$, $\sigma = 0.1$, and $\sigma = 0.2$.

SSSLRR-DWT denoising, an ablation study of SSSLRR-DWT is performed. SSSLRR-DWT is compared with the following denoising methods.

1) *SSWSR:* Only the ℓ_1 penalty is used, i.e., the cost function is

$$\widehat{\mathbf{W}} = \arg\min_{\mathbf{W}} \ \|\mathbf{H} - \mathbf{W}\|_F^2 + \lambda_1 \|\mathbf{W}\|_{1,1}. \tag{18}$$

2) *SSWLRR1:* Only the nuclear norm penalty is used, i.e., the cost function is

$$\widehat{\mathbf{W}} = \arg\min_{\mathbf{W}} \ \|\mathbf{H} - \mathbf{W}\|_F^2 + \lambda_* \|\mathbf{W}\|_* \tag{19}$$

where $\|\cdot\|_*$ is the nuclear norm.

3) *SSWLRR2:* The weighted nuclear norm low-rank penalty is used in the signal domain instead of using it in the wavelet domain, i.e., the cost function is

$$\widehat{\mathbf{X}} = \arg\min_{\mathbf{X}} \ \|\mathbf{Y} - \mathbf{X}\|_F^2 + \|\mathbf{X}\|_{\omega,*}. \tag{20}$$

4) *SLRR:* The ℓ_1 penalty and the weighted nuclear norm low-rank penalty are used in the signal domain, i.e., the cost function is

$$\widehat{\mathbf{X}} = \arg\min_{\mathbf{X}} \ \|\mathbf{Y} - \mathbf{X}\|_F^2 + \lambda_1 \|\mathbf{X}\|_{1,1} + \|\mathbf{X}\|_{\omega,*}. \tag{21}$$

The ADMM algorithm is adapted to get the solutions for these optimization problems.

The simulated WDCM dataset for Cases 1–3 and the UP dataset for Case 7 (the detailed experimental setting is given in Section III) are used for this experiment. Tables I and II and Figs. 2–8 present the denoising results. SSWSR can remove most of the Gaussian and the salt and pepper noise but can not remove the stripe noise. The difference in results between SSWLRR1 and SSWLRR2 shows that the extra flexibility provided by the weighted nuclear norm compared to the traditional nuclear norm leads to better denoising results. Both SLRR and SSSLRR-DWT can effectively remove mixed noise. The benefit of using the ℓ_1 and weighted nuclear norm

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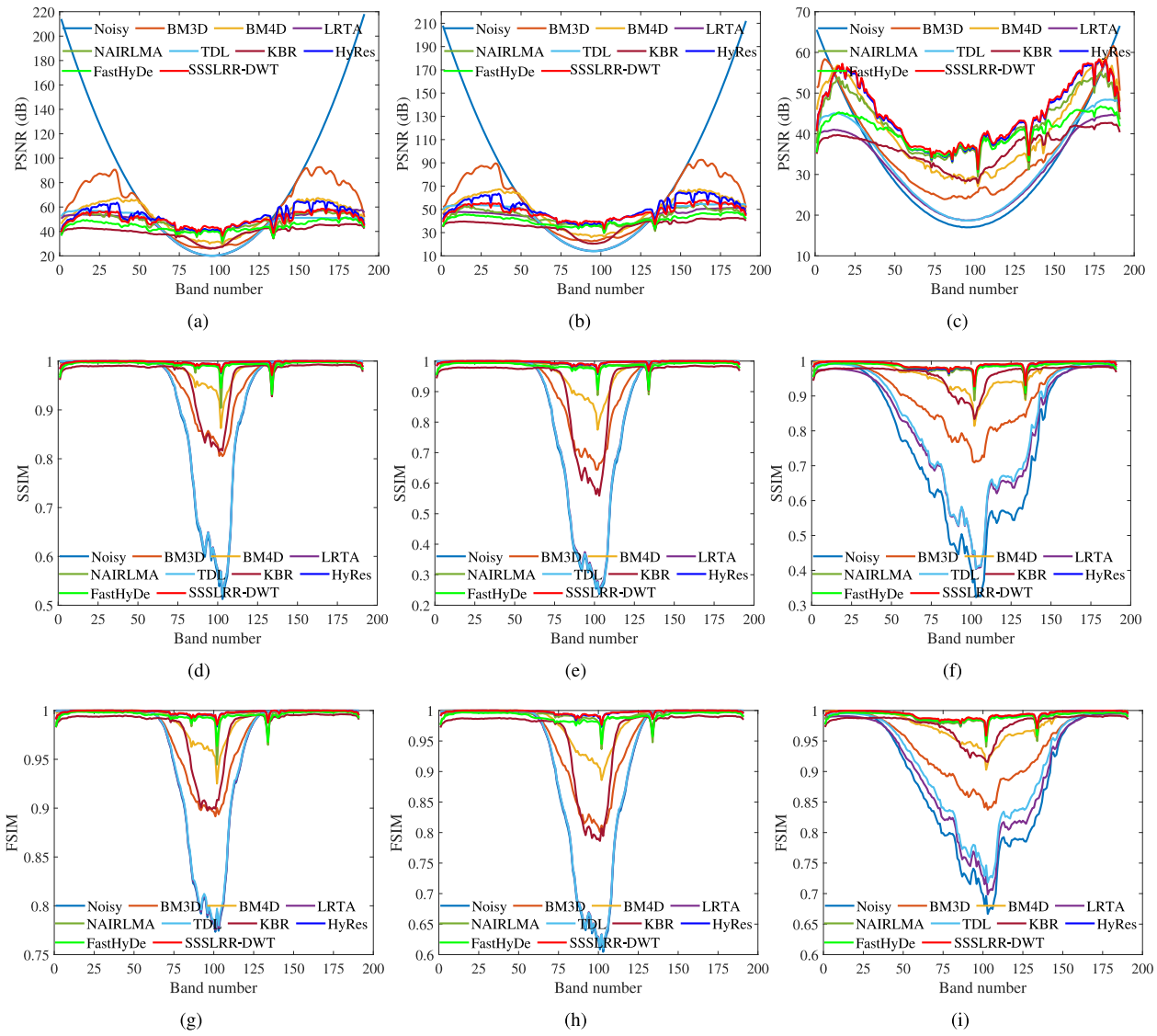


Fig. 16. (a)–(c) PSNR, (d)–(f) SSIM, and (g)–(i) FSIM values of each band of the denoised WDCM dataset. The first-to-third columns correspond to Case 2 with $\sigma = 0.5$, $\eta = 10$, $\sigma = 1$, $\eta = 10$, and $\sigma = 1$, $\eta = 20$.

low-rank penalties can clearly be seen by noticing that both SLRR and SSSLRR-DWT outperform SSWSR, SSWLRR1, and SSWLRR2. Finally, SSSLRR-DWT outperforms SLRR and yields the best results of all the methods, highlighting that working in the wavelet domain yields some gains.

III. Experimental Results and Discussion

In the experiments, both simulated and real datasets are used to demonstrate the effectiveness of the proposed method. Experimental results for SSSLRR-DWT are compared with the following denoising methods: two patch-based methods, i.e., BM3D[1] [15] and nonlocal transform-domain filter for volumetric data (BM4D[2]) [54], two low-rank-based methods, i.e., low-rank tensor approximation (LRTA[3]) [55] and noise-adjusted iterative low-rank matrix approximation (NAILRMA[4]) [34], four sparse and low-rank-based methods, i.e., tensor learning dictionary (TDL[5]) [56], Kronecker-basis-representation based tensor sparsity measure (KBR[6]) [57], sparse and low-rank model HSI restoration (HyRes[7]) [40], and fast hyperspectral denoising based on the low-rank and sparse representation (FastHyDe[8]) [39]. Each dataset is normalized between [0, 1] band by band before adding the noise and is returned to the original level after denoising.

A. Simulated Datasets Experiments

1) Data Description: We use two public HSI datasets to simulate noisy HSI. The first dataset is the Washington DC Mall (WDCM) dataset,[9] which was collected

[1]http://www.cs.tut.fi/~foi/GCF-BM3D/
[2]http://www.cs.tut.fi/~foi/GCF-BM3D/
[3]https://www.sandia.gov/tgkolda/TensorToolbox/
[4]https://sites.google.com/site/rshewei/home
[5]http://gr.xjtu.edu.cn/web/dymeng/3
[6]http://gr.xjtu.edu.cn/web/dymeng/3
[7]https://www.researchgate.net/profile/Behnood-Rasti
[8]https://github.com/LinaZhuang/FastHyDe_FastHyIn
[9]http://lesun.weebly.com/hyperspectral-dataset.html

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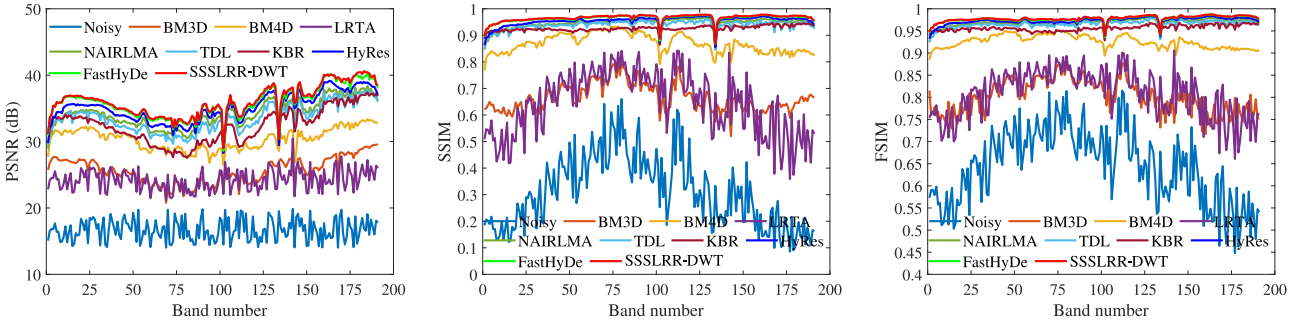


Fig. 17. (a) PSNR, (b) SSIM, and (c) FSIM values of each band of denoised WDCM dataset in Case 3 with $\sigma \sim U(0.1, 0.2)$.

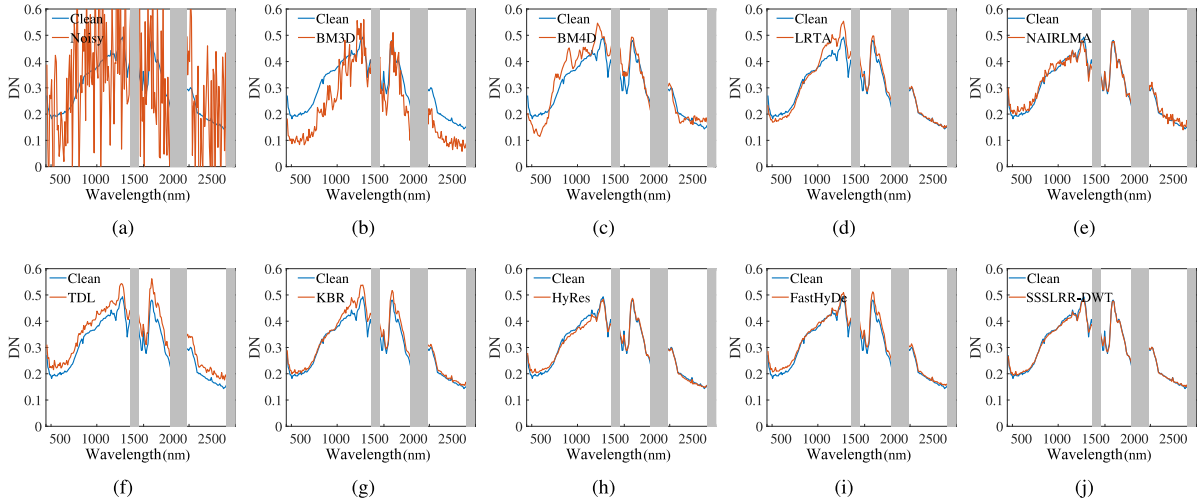


Fig. 18. Spectrum of pixel (173, 150) for the WDCM dataset, Case 1 with $\sigma = 0.2$, using different denoising methods. (a) Noisy image, (b) BM3D, (c) BM4D, (d) LRTA, (e) NAIRLMA (f), TDL, (g) KBR, (h) HyRes, (i) FastHyDe, and (j) SSSLRR-DWT. The water absorption bands are in gray.

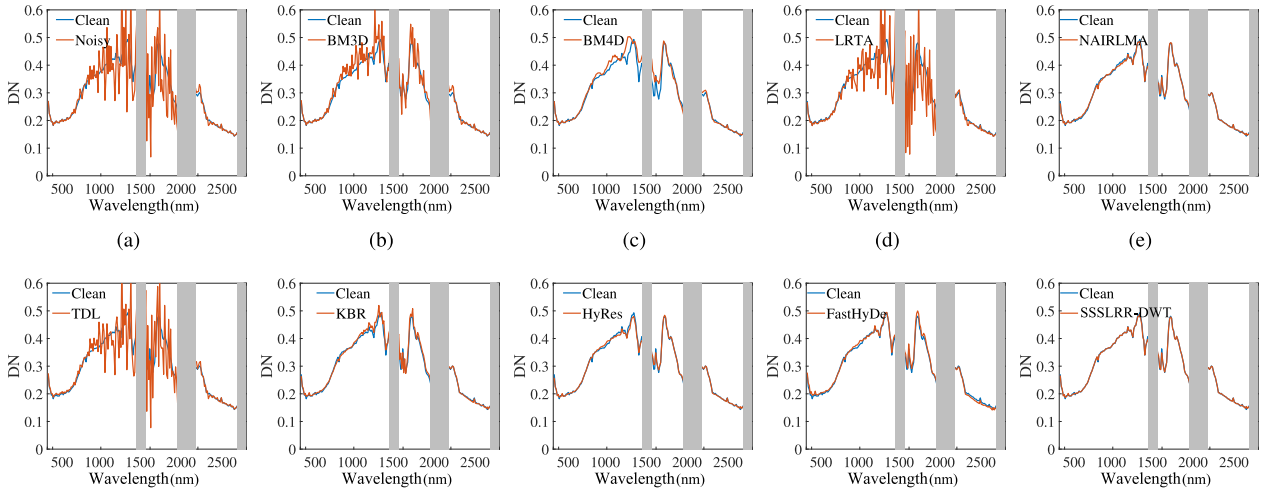


Fig. 19. Spectrum of pixel (173, 150) for the WDCM dataset, Case 2 with $\sigma = 1$, $\eta = 20$, using different denoising methods. (a) Noisy image, (b) BM3D, (c) BM4D, (d) LRTA, (e) NAIRLMA (f), TDL, (g) KBR, (h) HyRes, (i) FastHyDe, and (j) SSSLRR-DWT. The water absorption bands are in gray.

by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor and the whole image contains 1208 × 307 pixels with a spatial resolution of 2 m and 191 spectral bands. The second dataset is the University of Pavia (UP) dataset,[10] which was captured by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor and has 610 × 340 pixels with a spatial resolution of 1.3 m and 103 spectral bands. In the experiments, two small parts of the WDCM and UP datasets are used. Their sizes are 200 × 200 × 191 for the WDCM dataset and 200 × 200 × 103 for the UP dataset. Both datasets are assumed to be noise-

[10]http://lesun.weebly.com/hyperspectral-dataset.html

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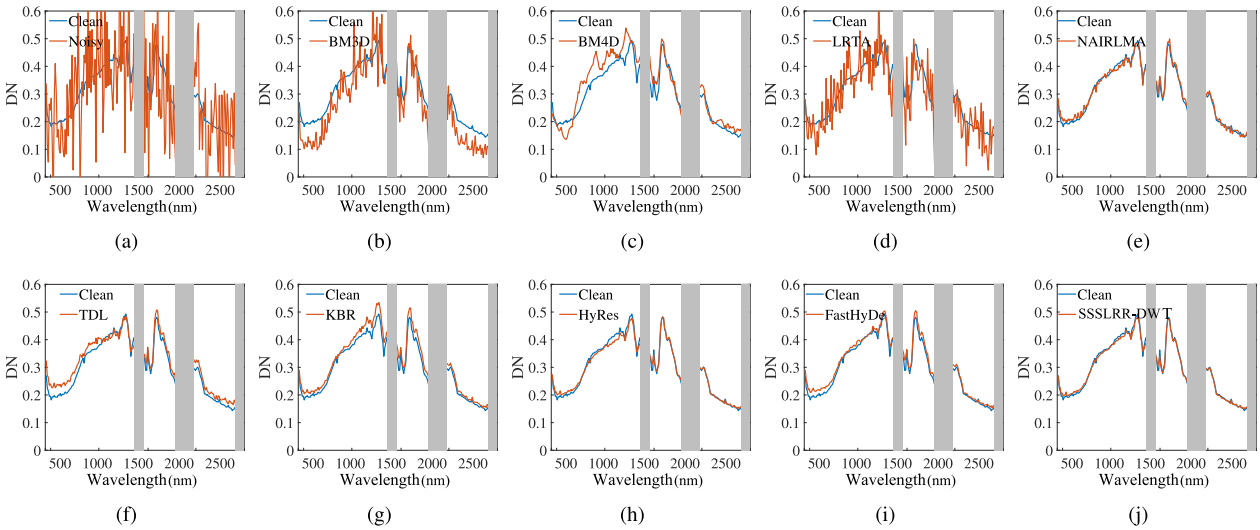


Fig. 20. Spectrum of pixel (173, 150) for the WDCM dataset, Case 3 with $\sigma \sim U(0.1, 0.2)$, using different denoising methods. (a) Noisy image, (b) BM3D, (c) BM4D, (d) LRTA, (e) NAIRLMA (f), TDL, (g) KBR, (h) HyRes, (i) FastHyDe, and (j) SSSLRR-DWT. The water absorption bands are in gray.

TABLE IV

QUANTITATIVE EVALUATION RESULTS [PSNR (dB), MSSIM, MFSIM, SAM (DEGREES), AND TIME (SECOND)] FOR THE DIFFERENT DENOISING METHODS USING THE UP DATASET. THE BEST RESULTS ARE IN BOLD TYPEFACE

UP	Noise level	Metric	Noisy	BM3D	BM4D	LRTA	NAIRLMA	TDL	KBR	HyRes	FastHyDe	SSSLRR-DWT
Case 1	$\sigma = 0.05$	PSNR	26.026	31.713	37.118	35.031	37.410	38.001	37.499	37.860	38.379	**38.807**
		MSSIM	0.6512	0.8931	0.9607	0.9304	0.9623	0.9680	0.9658	0.9646	0.9706	**0.9728**
		MFSIM	0.8549	0.9330	0.9784	0.9714	0.9837	0.9857	0.9831	0.9849	0.9862	**0.9875**
		SAM	13.900	6.1373	3.6802	4.6048	3.5091	3.1956	3.1937	3.3337	3.1292	**3.0070**
		Time	-	60.969	245.00	2.5617	27.388	4.9310	647.12	0.7363	**0.3052**	4.2892
	$\sigma = 0.1$	PSNR	20.002	28.063	33.603	31.461	33.812	34.526	34.803	34.332	35.545	**35.704**
		MSSIM	0.3880	0.7875	0.9234	0.8657	0.9165	0.9411	0.9457	0.9290	0.9507	**0.9520**
		MFSIM	0.7053	0.8659	0.9550	0.9342	0.9658	0.9706	0.9702	0.9681	0.9741	**0.9751**
		SAM	25.525	7.5487	5.0964	6.1013	5.0108	4.3931	3.9783	4.6130	4.0236	**3.9670**
		Time	-	59.788	243.39	0.7045	20.610	4.4749	681.14	0.6542	**0.3430**	4.5737
	$\sigma = 0.2$	PSNR	13.980	25.117	29.652	27.924	29.540	30.917	30.940	30.696	32.544	**32.997**
		MSSIM	0.1700	0.6401	0.8354	0.7505	0.8136	0.8832	0.8878	0.8612	0.9166	**0.9232**
		MFSIM	0.5242	0.7623	0.9061	0.8719	0.9236	0.9378	0.9333	0.9336	0.9520	**0.9567**
		SAM	42.357	9.3165	7.2347	7.6180	7.3585	5.7785	5.2794	6.4479	5.1509	**4.9689**
		Time	-	60.677	242.48	0.7807	21.781	4.1931	1257.4	0.6370	**0.3394**	4.8983
Case 2	$\sigma = 0.5$, $\eta = 10$	PSNR	26.144	34.154	39.424	27.257	38.215	27.306	34.766	41.055	40.213	**41.202**
		MSSIM	0.7731	0.9375	0.9758	0.7879	0.9706	0.8010	0.9329	0.9804	0.9783	**0.9822**
		MFSIM	0.8928	0.9591	0.9852	0.8970	0.9848	0.9097	0.9671	0.9909	0.9891	**0.9914**
		SAM	13.663	4.4674	2.7893	12.075	3.1835	12.019	4.8530	2.1801	2.5442	**2.3102**
		Time	-	57.901	244.17	1.2375	49.585	5.2778	642.83	0.8239	**0.2925**	13.858
	$\sigma = 1$, $\eta = 10$	PSNR	20.135	31.272	35.481	22.042	35.150	21.472	29.689	37.044	36.854	**37.721**
		MSSIM	0.6385	0.8920	0.9475	0.6513	0.9484	0.6718	0.8530	0.9531	0.9588	**0.9648**
		MFSIM	0.8060	0.9295	0.9682	0.8160	0.9729	0.8305	0.9283	0.9796	0.9778	**0.9813**
		SAM	25.097	5.8888	4.1183	20.728	4.3844	22.023	8.7923	3.2496	3.4898	**3.2060**
		Time	-	56.629	244.10	3.6193	49.192	5.3541	639.44	0.7383	**0.3294**	13.436
	$\sigma = 1$, $\eta = 20$	PSNR	20.124	29.355	34.217	24.728	34.460	27.264	34.465	35.703	36.191	**36.484**
		MSSIM	0.4599	0.8331	0.9291	0.6189	0.9299	0.7401	0.9358	0.9411	0.9542	**0.9565**
		MFSIM	0.7313	0.8936	0.9575	0.8000	0.9682	0.8866	0.9656	0.9729	0.9756	**0.9773**
		SAM	25.205	7.0915	4.8462	15.492	4.7633	11.523	4.4364	3.9780	3.7887	**3.6863**
		Time	-	59.374	243.54	2.4560	29.093	5.8211	644.80	0.6525	**0.3370**	4.6615
Case 3	$\sigma \sim U$ (0.1,0.2)	PSNR	16.191	26.003	31.098	23.846	31.275	30.760	32.209	32.537	33.630	**34.270**
		MSSIM	0.2536	0.6879	0.8756	0.5575	0.8676	0.8516	0.9134	0.8990	0.9310	**0.9379**
		MFSIM	0.6008	0.8049	0.9275	0.7719	0.9456	0.9418	0.9496	0.9529	0.9614	**0.9660**
		SAM	35.723	9.1650	6.3804	16.364	6.2691	6.7892	4.8065	5.3489	4.7269	**4.4690**
		Time	-	62.464	243.46	2.0414	18.791	4.8703	1173.4	0.6298	**0.3412**	4.9486

free HSIs. The noisy simulated datasets are created as follows.

1) *Case 1:* A zero-mean isotropic Gaussian noise with standard deviation $\sigma \in \{0.05, 0.1, 0.2\}$ is added to each band.
2) *Case 2:* A zero-mean bandwise Gaussian noise with varied variance, σ_i, is added to each band. The bell shape variance varies according to

$$\sigma_i^2 = \sigma^2 \frac{e^{-\frac{(i-B/2)^2}{2\eta^2}}}{\sum_{j=1}^{B} e^{-\frac{(i-B/2)^2}{2\eta^2}}} \tag{22}$$

where σ and η control the noise intensity and the bell width, respectively [58]. We set σ and η to two different values 0.5 and 1, 10 and 20, respectively.

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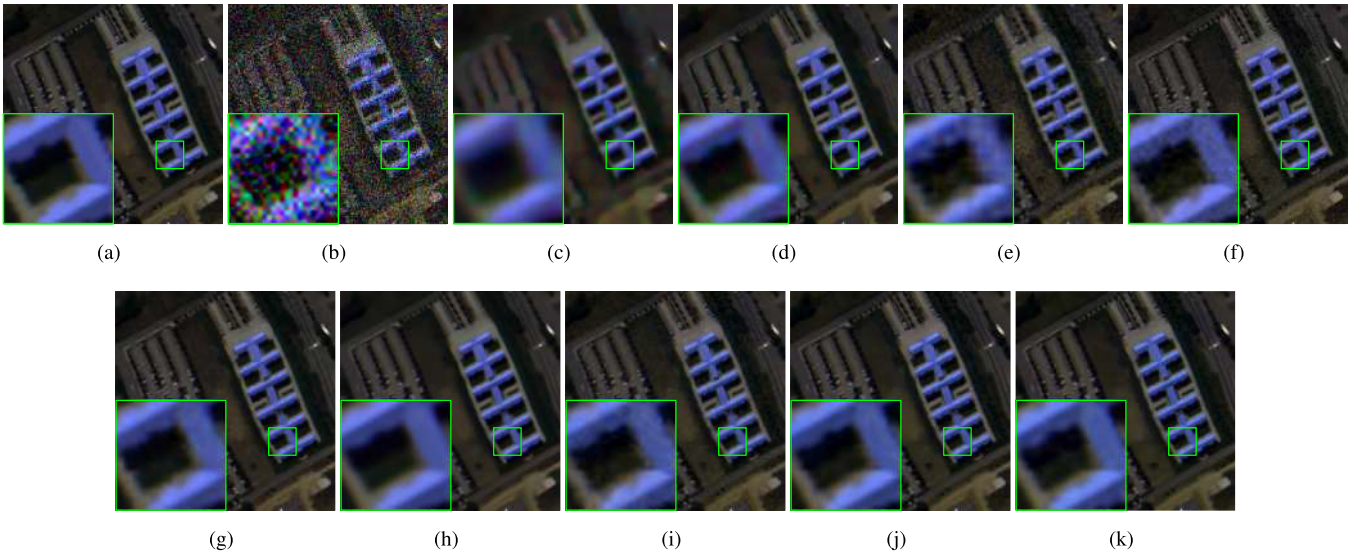


Fig. 21. Denoising for the UP dataset, Case 1 with $\sigma = 0.2$, using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 64, G: 45, and B: 10), (b) noisy image, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

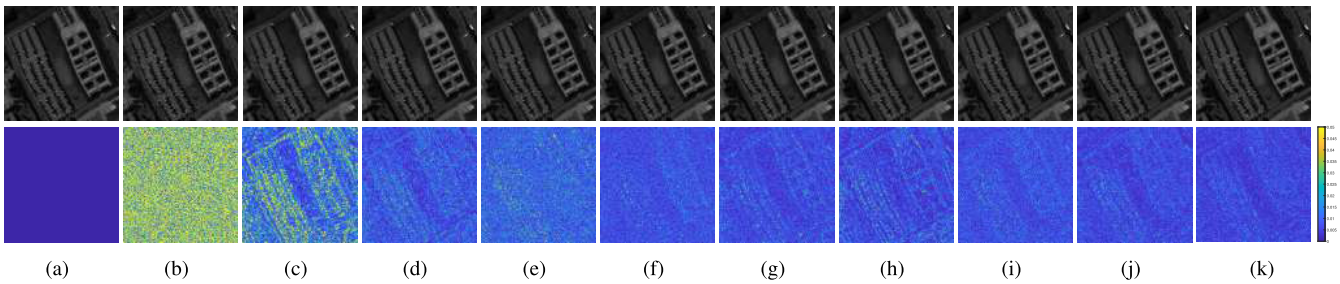


Fig. 22. Denoised images (the first row) and residual images (the second row) for the UP dataset, Case 1 with $\sigma = 0.05$, using different methods. (a) Original band 51, (b) noisy band 51, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

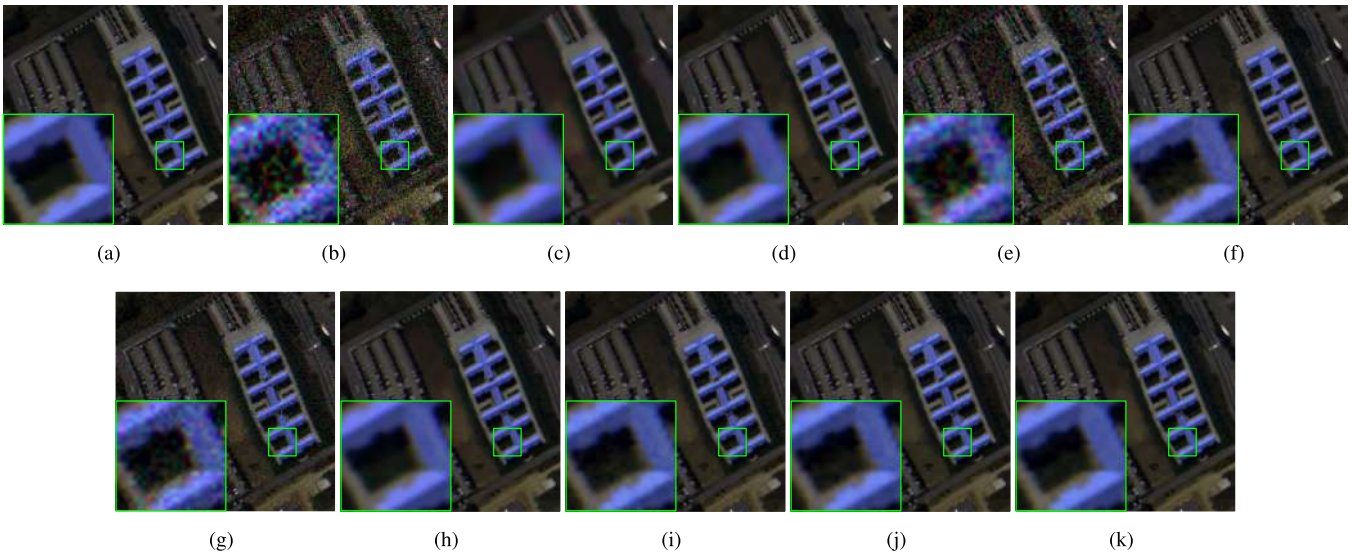


Fig. 23. Denoising for the UP dataset, Case 2 with $\sigma = 1$, $\eta = 20$ using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 64, G: 45, and B: 10), (b) noisy image, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

3) *Case 3:* A zero-mean Gaussian noise that is added to each band with standard deviation σ is drawn from a uniform distribution between 0.1 and 0.2 [$\sigma \sim U(0.1, 0.2)$].

2) Evaluation Metric: The quantitative metrics used to evaluate the results for the simulated datasets are the peak signal-to-noise ratio (PSNR), the mean structural

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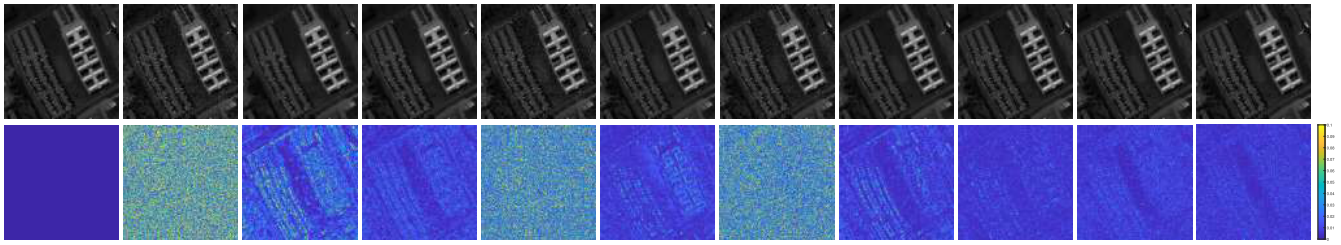


Fig. 24. Denoised images (the first row) and residual images (the second row) for the UP dataset, Case 2 with $\sigma = 1$, $\eta = 10$, using different methods. (a) Original band 30, (b) noisy band 30, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

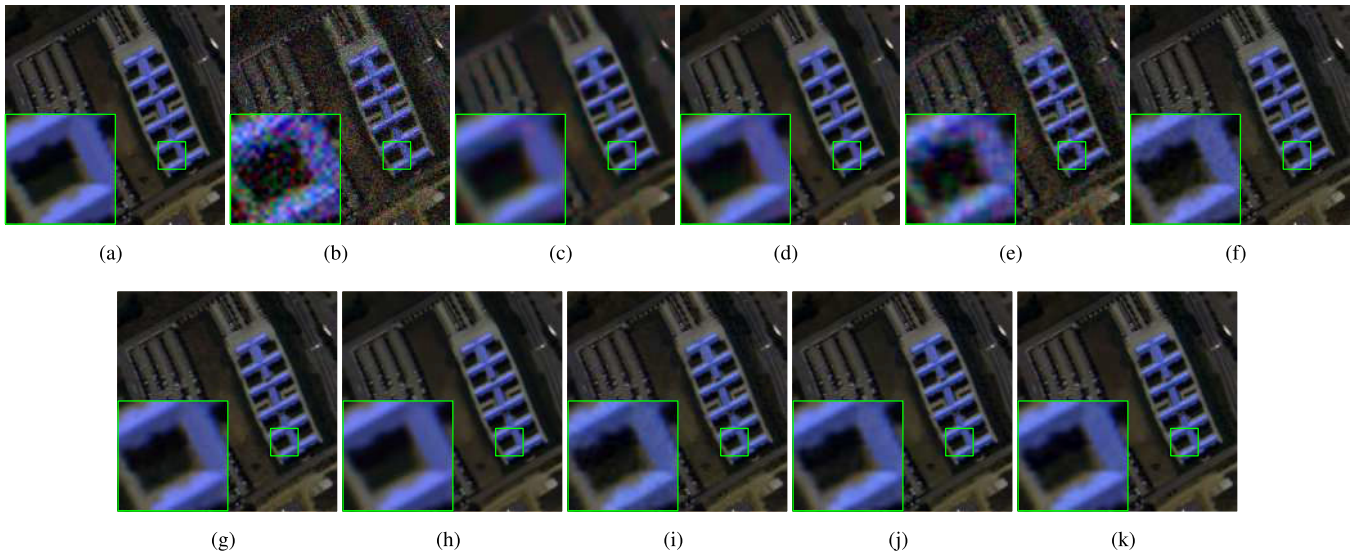


Fig. 25. Denoising for the UP dataset, Case 3 with $\sigma \sim U(0.1, 0.2)$, using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 64, G: 45, and B: 10), (b) noisy image, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

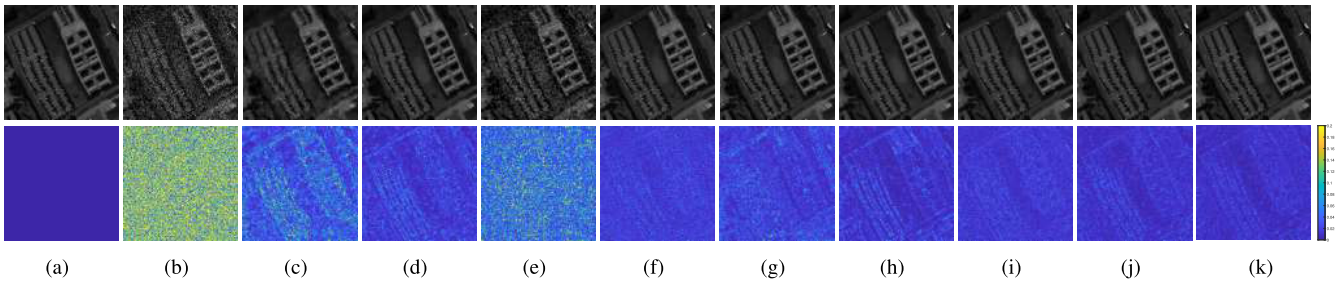


Fig. 26. Denoised images (the first row) and residual images (the second row) for the UP dataset, Case 3 with $\sigma \sim U(0.1, 0.2)$, using different methods. (a) Original band 51, (b) noisy band 51, (c) BM3D, (d) BM4D, (e) LRTA, (f) NAIRLMA, (g) TDL, (h), KBR, (i) HyRes, (j) FastHyDe, and (k) SSSLRR-DWT.

similarity (MSSIM), the mean feature similarity (MFSIM), and the spectral angle mapper (SAM) that are used. The definition of PSNR is

$$\text{PSNR} = 10\log_{10}\left(\frac{\max^2(\mathbf{x})}{\frac{1}{P}\parallel \mathbf{x} - \hat{\mathbf{x}} \parallel_2^2}\right) \tag{23}$$

where $\mathbf{x}$ is the clean image and $\hat{\mathbf{x}}$ is the denoised image. MSSIM is the bandwise mean of SSIM, and the SSIM of ith band is defined as [59]

$$\text{SSIM}_{i,j} = \frac{(2\mu_{x_{i,j}}\mu_{\hat{x}_{i,j}} + c_1)(2\sigma_{x_{i,j}\hat{x}_{i,j}} + c_2)}{\left(\mu_{x_{i,j}}^2 + \mu_{\hat{x}_{i,j}}^2 + c_1\right)\left(\sigma_{x_{i,j}}^2 + \sigma_{\hat{x}_{i,j}}^2 + c_2\right)} \tag{24}$$

where $\mu_{x_{i,j}}$, $\mu_{\hat{x}_{i,j}}$, $\sigma_{x_{i,j}}$, and $\sigma_{\hat{x}_{i,j}}$ are the mean and standard deviation for the reference and estimated images in a local 11×11 window centered at pixel i, respectively. $\sigma_{x_{i,j}\hat{x}_{i,j}}$ is the cross covariance between two images. $c_1 = (K_1 M)^2$ and $c_2 = (K_2 M)^2$, where $K_1 = 0.01$, $K_2 = 0.03$, and M is the

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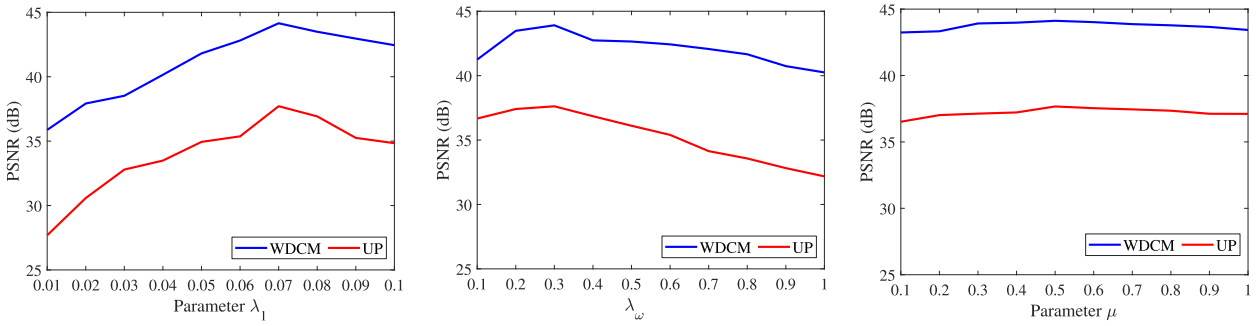


Fig. 27. PSNR as a function of the parameters (a) λ_1, (b) λ_ω (is a constant and is related to the weight ω_i), and (c) μ for SSSLRR-DWT denoising method for the WDCM and the UP datasets.

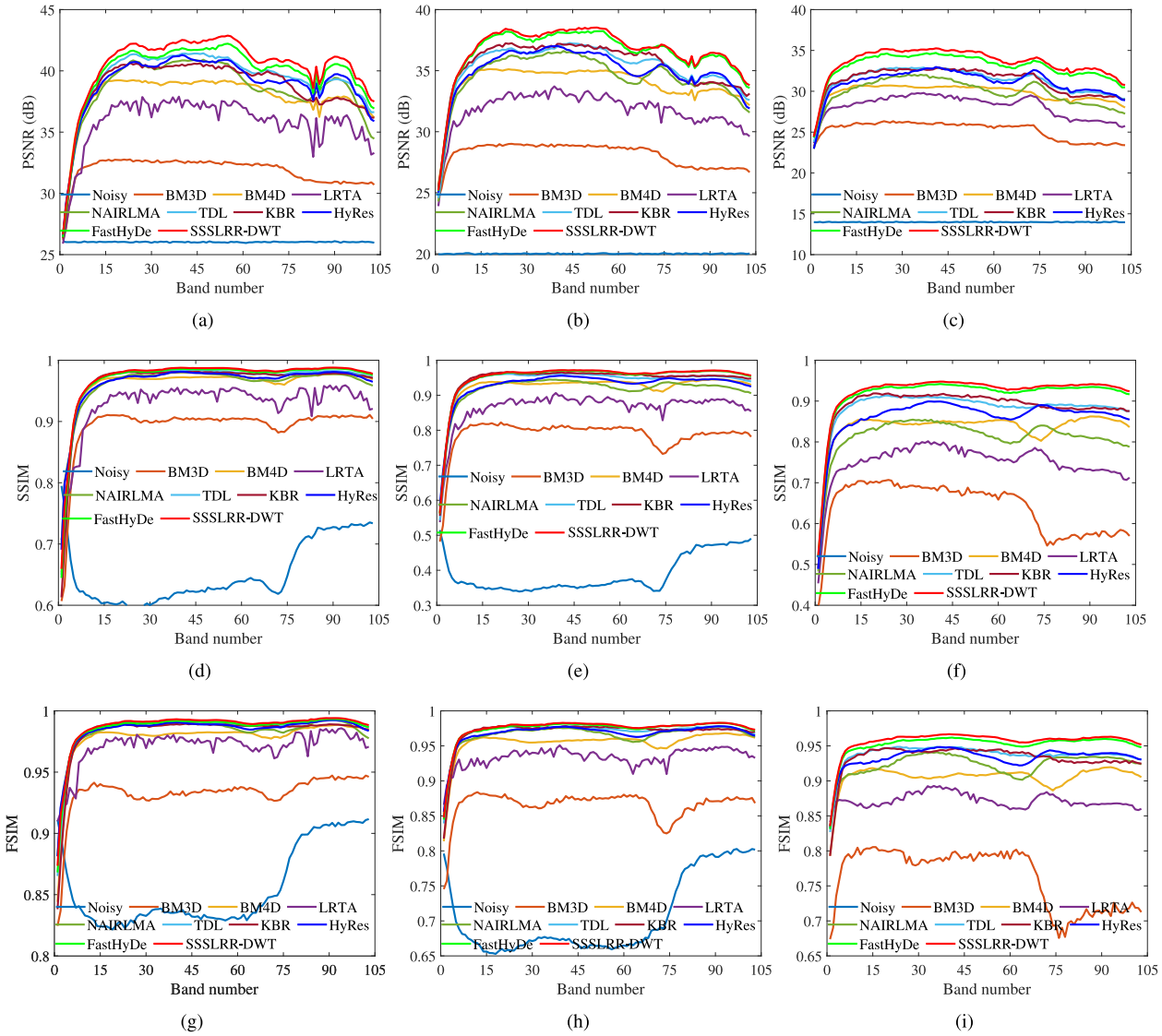


Fig. 28. (a)–(c) PSNR, (d)–(f) SSIM, and (g)–(i) FSIM values of each band of the denoised UP dataset. The first-to-third columns correspond to Case 1 with $\sigma = 0.05$, $\sigma = 0.1$, and $\sigma = 0.2$.

dynamical range of the image that is set to 1 in this article. PSNR and MSSIM assess the similarity of the denoised image and the clean image based on the mean square error and the structural consistency, respectively. MFSIM focuses on the perceptual consistency based on the phase congruency and gradient magnitude between the target image and the reference

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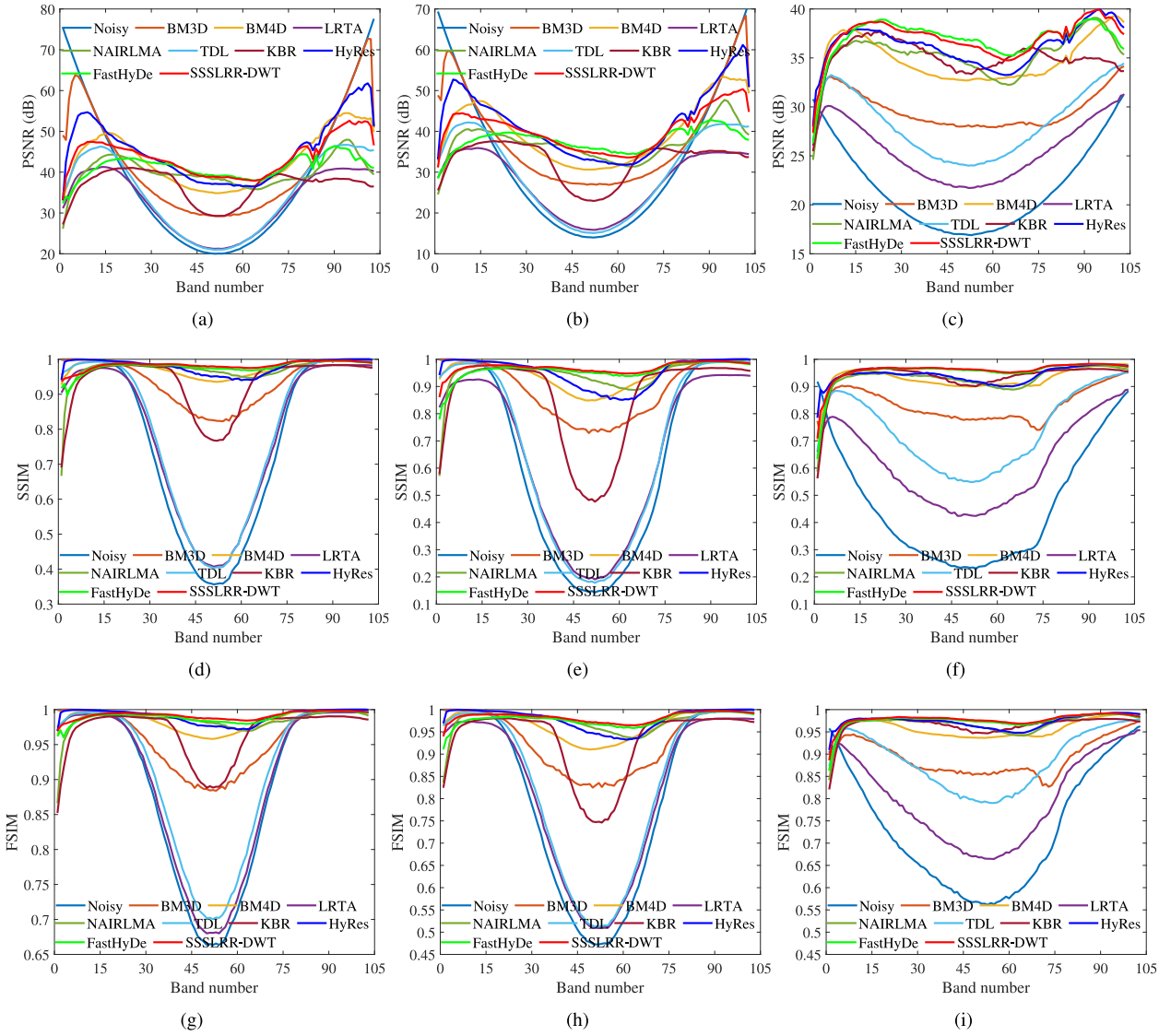


Fig. 29. (a)–(c) PSNR, (d)–(f) SSIM, and (g)–(i) FSIM values of each band of the denoised UP dataset. The first-to-third columns correspond to Case 2 with $\sigma = 0.5$, $\eta = 10$, $\sigma = 1$, $\eta = 10$, and $\sigma = 1$, $\eta = 20$.

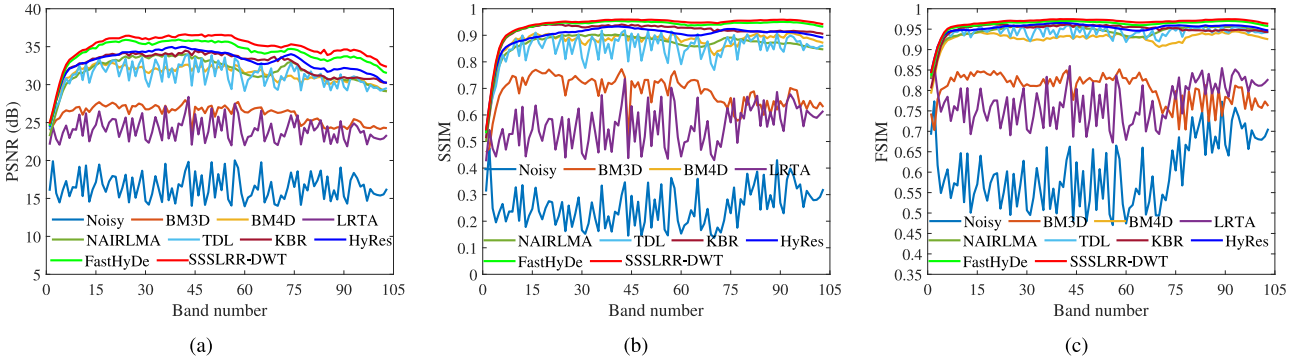


Fig. 30. (a) PSNR, (b) SSIM, and (c) FSIM values of each band of the denoised UP dataset in Case 3 with $\sigma \sim U(0.1, 0.2)$.

image [60]. The larger the PSNR, MSSIM, and FSIM are, the closer the denoised HSI is to the clean one. SAM calculates the mean angle between spectrum vectors of the target image and the reference image across all spatial positions. The definition of SAM in degrees is

$$\mathrm{SAM} = \frac{1}{nB}\sum_{p=1}^{nB}\arccos\left(\frac{\mathbf{x}_{(p)}^{T}\widehat{\mathbf{x}}_{(p)}}{\|\mathbf{x}_{(p)}\|\|\widehat{\mathbf{x}}_{(p)}\|}\right)\frac{180}{\pi} \tag{25}$$

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TABLE V

QUANTITATIVE EVALUATION RESULTS [PSNR (dB), MSSIM, MFSIM, SAM (DEGREES), AND TIME (SECOND)] FOR THE DIFFERENT DENOISING METHODS USING THE WASHINGTON DC MALL (WDCM) DATASET. THE BEST RESULTS ARE IN BOLD TYPEFACE

WDCM	Noise level	Metric	Noisy	LRMR	Ngmeet	TLR-ℓ_0TV	SSSLRR-DCT	SSSLRR-DWT
Case 1	$\sigma = 0.05$	PSNR	26.025	38.529	41.453	39.925	41.546	**41.681**
		MSSIM	0.7191	0.9802	**0.9910**	0.9858	0.9907	0.9908
		MFSIM	0.8691	0.9879	**0.9942**	0.9913	0.9940	**0.9942**
		SAM	18.048	4.6330	2.9266	3.6415	2.9036	**2.8475**
		Time	-	24.506	38.838	1011.4	11.836	**11.754**
	$\sigma = 0.1$	PSNR	20.004	34.082	**38.089**	35.572	37.739	37.968
		MSSIM	0.4528	0.9473	**0.9805**	0.9655	0.9785	0.9794
		MFSIM	0.7348	0.9700	**0.9878**	0.9789	0.9861	0.9867
		SAM	30.350	7.3018	**3.8188**	5.4473	4.1484	4.0887
		Time	-	25.062	39.104	1031.1	12.882	**12.593**
	$\sigma = 0.2$	PSNR	13.983	29.687	33.990	33.277	34.015	**34.518**
		MSSIM	0.2061	0.8734	0.9544	0.9516	0.9547	**0.9582**
		MFSIM	0.5644	0.9335	0.9717	0.9697	0.9721	**0.9734**
		SAM	46.309	11.069	5.8062	5.9847	5.8048	**5.6292**
		Time	-	24.571	48.150	1035.6	13.659	**13.466**
Case 2	$\sigma = 0.5$, $\eta = 10$	PSNR	28.840	38.094	39.511	43.957	45.966	**46.750**
		MSSIM	0.9321	0.9825	0.9874	0.9962	0.9943	**0.9979**
		MFSIM	0.9641	0.9886	0.9923	0.9971	0.9975	**0.9985**
		SAM	13.704	5.0619	3.6387	2.3002	1.9318	**1.5664**
		Time	-	**21.054**	35.629	1009.3	44.387	44.097
	$\sigma = 1$, $\eta = 10$	PSNR	22.820	34.332	33.472	41.545	43.727	**44.165**
		MSSIM	0.8674	0.9642	0.9655	0.9932	0.9961	**0.9968**
		MFSIM	0.9306	0.9778	0.9777	0.9942	0.9975	**0.9977**
		SAM	23.863	7.4990	7.0825	2.8441	2.1047	**1.9880**
		Time	-	**23.543**	37.961	1012.8	44.229	44.075
	$\sigma = 1$, $\eta = 20$	PSNR	22.815	35.041	34.315	39.135	40.198	**40.436**
		MSSIM	0.7853	0.9670	0.9691	0.9885	0.9910	**0.9913**
		MFSIM	0.8927	0.9789	0.9809	0.9914	0.9934	**0.9939**
		SAM	23.972	6.7136	6.9198	3.4272	3.1831	**3.0618**
		Time	-	**25.728**	39.083	1016.6	37.125	36.186
Case 3	$\sigma \sim U(0.1, 0.2)$	PSNR	16.371	31.519	34.364	34.672	35.435	**35.544**
		MSSIM	0.3083	0.9112	0.9573	0.9549	0.9659	**0.9661**
		MFSIM	0.6423	0.9518	0.9734	0.9738	0.9781	**0.9782**
		SAM	39.689	9.3786	6.2681	6.1109	5.2643	**5.2188**
		Time	-	24.746	37.458	1014.6	13.418	**12.620**

where $\mathbf{x}_{(p)}$ and $\widehat{\mathbf{x}}_{(p)}$ are the clean image and denoised image at pixel p, respectively. Unlike the former three metrics, the smaller the SAM is, the better the results are.

3) Parameters Setting: All the parameters used for the comparison methods follow either the authors' source code settings or are experimentally evaluated according to the reference recommendations. SSSLRR-DWT uses three parameters, and they are set as follows for all the simulated and real datasets experiments, the regularization parameters λ_1 and μ are set to 0.07 and 0.5, respectively, and λ_ω is a constant, is related to the weight ω_i, and is set to 0.3. We give the reason for this decision in the discussion part.

4) Results on WDCM Dataset: Table III presents the PSNR, MSSIM, MFSIM, SAM, and running time of SSSLRR-DWT and all comparison methods for the WDCM dataset. The highest values of PSNR, MSSIM, and MFSIM and the lowest values of SAM and running time are shown in bold. SSSLRR-DWT outperforms other methods in terms of all metrics. Also, SSSLRR-DWT yields the best performance for all noise settings. Specifically, SSSLRR-DWT achieves an approximately 0.6 dB (Case 1 and 3) and 5 dB (Case 2), 0.5 dB (Case 2), and 2 dB (Case 1 and 3) improvement in PSNR, and a few percent of MSSIM and MFSIM in comparison with the second-best methods, FastHyDe and HyRes. Figs. 15–17 show the PSNR, SSIM, and FSIM for each band in Cases 1–3 for the WDCM dataset. SSSLRR-DWT (red curve) has the best PSNR, SSIM, and FSIM for almost all the bands. This also shows the effectiveness of SSSLRR-DWT for HSI denoising. Figs. 18–20 show the spectral signatures of pixel (173, 150) for the WDCM dataset with different noise settings. From Figs. 18–20, it can be clearly observed that the proposed SSSLRR-DWT yields a higher similarity of the spectral signatures with the clean pixel than the other denoising methods.

In terms of visual quality, Figs. 9, 11, and 13 show the pseudocolor images of the WDCM dataset denoised by nine

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TABLE VI

QUANTITATIVE EVALUATION RESULTS [PSNR (dB), MSSIM, MFSIM, SAM (DEGREES), AND TIME (SECOND)] FOR THE DIFFERENT DENOISING METHODS USING THE WASHINGTON DC MALL (WDCM) DATASET. THE BEST RESULTS ARE IN BOLD TYPEFACE

Noise level	Metric	Noisy	LRMR	NGmeet	TLR-ℓ_0TV	SSSLRR-DCT	SSSLRR-DWT
Case 4	PSNR	11.376	28.233	29.194	29.613	32.381	**32.490**
	MSSIM	0.1831	0.8357	0.8809	0.8826	0.9321	**0.9345**
	MFSIM	0.5228	0.9162	0.9317	0.9338	0.9591	**0.9599**
	SAM	53.557	12.914	7.5161	8.8449	7.1005	**6.7911**
	Time	-	30.622	44.436	1135.7	12.793	**12.586**
Case 5	PSNR	12.829	29.356	28.571	29.787	31.633	**32.053**
	MSSIM	0.1792	0.8766	0.8633	0.8886	0.9241	**0.9251**
	MFSIM	0.5766	0.9432	0.9206	0.9350	0.9532	**0.9562**
	SAM	43.896	9.4171	9.3268	8.6681	7.4483	**7.2766**
	Time	-	30.191	44.701	1143.3	12.873	**12.191**
Case 6	PSNR	16.370	29.419	26.798	30.266	33.502	**33.851**
	MSSIM	0.3104	0.8935	0.796	0.8988	0.9433	**0.9515**
	MFSIM	0.6432	0.9453	0.8828	0.9416	0.9665	**0.9692**
	SAM	39.813	9.8694	8.0446	8.4550	6.5269	**6.0191**
	Time	-	29.913	43.974	1133.5	12.491	**12.448**
Case 7	PSNR	12.698	27.659	24.205	28.341	30.636	**31.309**
	MSSIM	0.1714	0.8483	0.8208	0.8563	0.9077	**0.9139**
	MFSIM	0.5694	0.9318	0.9388	0.9175	0.9439	**0.9501**
	SAM	44.373	10.215	11.504	9.7621	7.8932	**7.8694**
	Time	-	30.145	44.338	1138.2	12.752	**12.381**

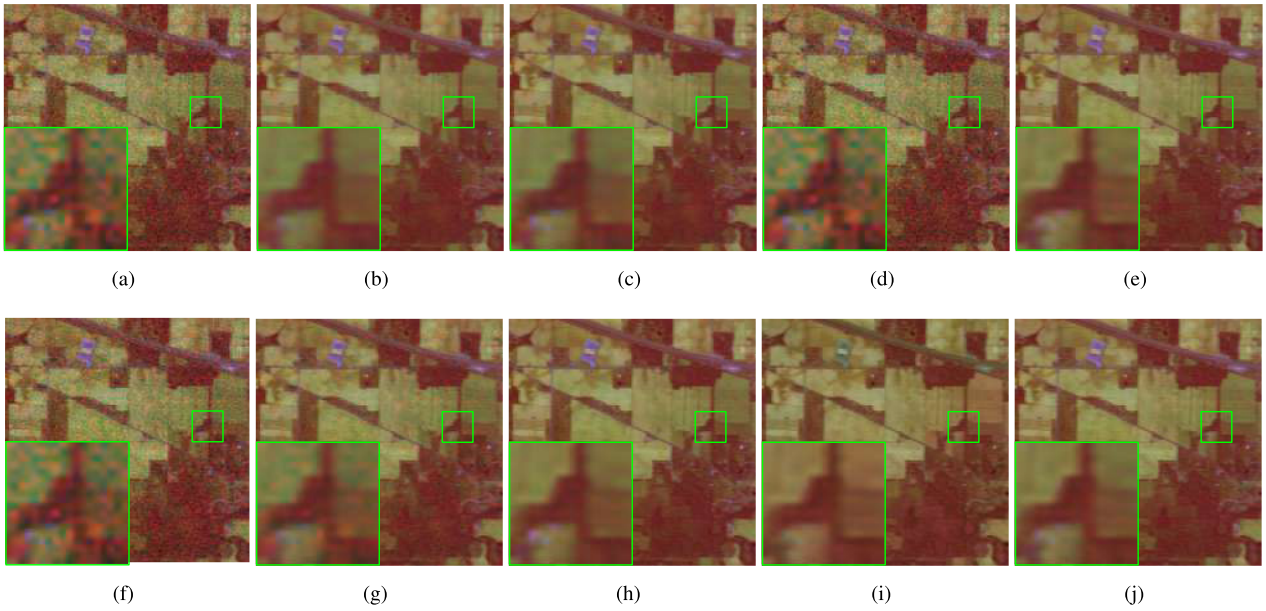


Fig. 31. Denoising for the Indian Pines dataset. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 220, G: 144, and B: 3), (b) BM3D, (c) BM4D, (d) LRTA, (e) NAIRLMA (f), TDL, (g) KBR, (h) HyRes, (i) FastHyDe, and (j) SSSLRR-DWT.

different methods under Case 1 with $\sigma = 0.2$, Case 2 with $\sigma = 1$, $\eta = 20$, and Case 3 with $\sigma \sim U(0.1, 0.2)$, respectively. The same area of each subfigure marked with a green box was enlarged for a better comparison. For LRTA, there is still noise in the denoised images, as can be seen in Figs. 9, 11, and 13. BM3D removes the noise but gives oversmooth denoised images. All the remaining methods effectively remove the noise. It is visually difficult to compare the comparison denoising methods. To show the denoising results, the difference between each original band and the denoised band is used, i.e. residual image. Figs. 10, 12, and 14 show band 95 of the denoising results and its residual images under Case 1 with $\sigma = 0.05$, Case 2 with $\sigma = 1$ and $\eta = 10$, and Case 3 with $\sigma \sim U(0.1, 0.2)$, respectively. By comparing the residual images, SSSLRR-DWT achieves the smallest residual in all cases. NAIRLMA, HyRes, and FastHyDe also do good job in denoising in most cases and achieve small residual. However, for the denoising WDCM dataset in Case 2 (see Fig. 12), NAIRLMA and FastHyDe are slightly worse than HyRes and SSSLRR-DWT.

5) UP Dataset: Table IV presents the PSNR, MSSIM, MFSIM, SAM, and running time of SSSLRR-DWT and all comparison denoising methods for the UP dataset. The highest values of PSNR, MSSIM, and MFSIM and the lowest values of SAM and running time are highlighted in bold. It can be seen that the SSSLRR-DWT

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TABLE VII

QUANTITATIVE EVALUATION RESULTS [PSNR (dB), MSSIM, MFSIM, SAM (DEGREES), AND TIME (SECOND)] FOR THE DIFFERENT DENOISING METHODS USING THE UP DATASET. THE BEST RESULTS ARE IN BOLD TYPEFACE

UP	Noise level	Metric	Noisy	LRMR	Ngmeet	TLR-ℓ_0TV	SSSLRR-DCT	SSSLRR-DWT
Case 1	$\sigma = 0.05$	PSNR	26.026	36.588	**39.090**	37.1927	38.143	38.807
		MSSIM	0.6512	0.9538	**0.9772**	0.9611	0.9682	0.9728
		MFSIM	0.8549	0.9802	**0.9891**	0.9834	0.9852	0.9875
		SAM	13.900	3.9784	**2.7183**	3.5889	3.2753	3.0070
		Time	-	22.326	35.553	966.89	4.3200	**4.2892**
	$\sigma = 0.1$	PSNR	20.002	32.279	**36.173**	34.131	35.356	35.704
		MSSIM	0.3880	0.8862	**0.9611**	0.9288	0.9514	0.9520
		MFSIM	0.7053	0.9535	**0.9804**	0.9678	0.9745	0.9751
		SAM	25.525	6.2864	**3.4468**	4.7002	4.2018	3.9670
		Time	-	23.732	37.725	971.42	4.8847	**4.5737**
	$\sigma = 0.2$	PSNR	13.980	27.753	32.484	31.681	32.605	**32.997**
		MSSIM	0.1700	0.7473	0.9152	0.9088	0.9164	**0.9232**
		MFSIM	0.5242	0.8965	0.9509	0.9428	0.9514	**0.9567**
		SAM	42.357	9.8416	5.2406	5.7382	5.2012	**4.9689**
		Time	-	24.493	43.302	968.23	4.9211	**4.8983**
Case 2	$\sigma = 0.5$, $\eta = 10$	PSNR	26.144	35.085	35.367	39.110	40.472	**41.202**
		MSSIM	0.7731	0.9348	0.9532	0.9746	0.9788	**0.9822**
		MFSIM	0.8928	0.9712	0.9770	0.9842	0.9894	**0.9914**
		SAM	13.663	4.9504	4.6545	2.8983	2.5342	**2.3102**
		Time	-	23.433	35.457	971.22	14.682	**13.858**
	$\sigma = 1$, $\eta = 10$	PSNR	20.135	31.435	29.585	34.948	37.141	**37.721**
		MSSIM	0.6385	0.8750	0.9011	0.9566	0.9567	**0.9648**
		MFSIM	0.8060	0.9452	0.9511	0.9687	0.9803	**0.9813**
		SAM	25.097	7.3800	9.0351	4.0144	3.4374	**3.2060**
		Time	-	24.753	35.889	967.97	13.491	**13.436**
	$\sigma = 1$, $\eta = 20$	PSNR	20.124	32.279	32.304	35.073	36.320	**36.484**
		MSSIM	0.4599	0.8821	0.9091	0.9387	0.9557	**0.9565**
		MFSIM	0.7313	0.9505	0.9578	0.9675	0.9761	**0.9773**
		SAM	25.205	6.4972	6.6344	4.2612	3.7267	**3.6863**
		Time	-	24.339	36.044	968.80	8.7625	**4.6615**
Case 3	$\sigma \sim U(0.1, 0.2)$	PSNR	16.191	29.575	33.357	33.416	34.164	**34.270**
		MSSIM	0.2536	0.8147	0.9301	0.9208	0.9372	**0.9379**
		MFSIM	0.6008	0.9246	0.9637	0.9624	0.9656	**0.9660**
		SAM	35.723	8.2789	5.2755	4.8358	4.4698	**4.4690**
		Time	-	23.236	33.619	968.21	6.5452	**4.9486**

gets the best results in terms of all metrics. SSSLRR-DWT achieves an approximately 0.5–2 dB gain in PSNR. This further validates that our proposed SSSLRR-DWT method can remove noise effectively. Figs. 28–30 show the PSNR, SSIM, and FSIM of each band in Cases 1–3 for the UP dataset. Compared with all the other methods, SSSLRR-DWT achieves higher PSNR, SSIM, and FSIM (red curve) for almost all the bands, showing the robustness of SSSLRR-DWT for HSI denoising.

Figs. 21, 23, and 25 show the pseudocolor images of the UP dataset denoised by nine different methods under Case 1 with $\sigma = 0.2$, Case 2 with $\sigma = 1$, $\eta = 20$, and Case 3 with $\sigma \sim U(0.1, 0.2)$, respectively. The same area of each subfigure marked with a green box is enlarged for a better comparison. As shown in Figs. 21, 23, and 25, BM3D smooths as before some details information of the original image, and LRTA still cannot completely remove the noise. All the other methods get satisfactory denoising results in most cases. Figs. 22, 24, and 26 show band 51 of the denoising results and its residual images under Case 1 with $\sigma = 0.05$, Case 2 with $\sigma = 1$, $\eta = 10$, and Case 3 with $\sigma \sim U(0.1, 0.2)$, respectively. As observed, the proposed SSSLRR-DWT method achieves good denoising result.

B. Real Datasets Experiments

1) Data Description: The Indian Pines and the Urban datasets are used in the real datasets experiments. The Indian Pines dataset[11] was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor and the image contains 145×145 pixels and 220 spectral bands. The Urban dataset[12] was collected by HYDICE and contains 307×307 pixels and 210 spectral bands. Both Indian Pines and Urban datasets are

[11] http://lesun.weebly.com/hyperspectral-dataset.html
[12] http://lesun.weebly.com/hyperspectral-dataset.html

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TABLE VIII

QUANTITATIVE EVALUATION RESULTS [PSNR (dB), MSSIM, MFSIM, SAM (DEGREES), AND TIME (SECOND)] FOR THE DIFFERENT DENOISING METHODS USING THE UP DATASET. THE BEST RESULTS ARE IN BOLD TYPEFACE

Noise level	Metric	Noisy	LRMR	NGmeet	TLR-ℓ_0TV	SSSLRR-3DCT	SSSLRR-DWT
Case 4	PSNR	11.645	26.204	26.089	29.261	32.477	**32.567**
	MSSIM	0.1530	0.6893	0.7831	0.8032	0.9201	**0.9216**
	MFSIM	0.4891	0.8707	0.8999	0.9199	0.9587	**0.9607**
	SAM	49.501	11.907	12.345	8.5770	5.8926	**5.8172**
	Time	-	30.341	44.308	971.14	6.8324	**5.7550**
Case 5	PSNR	12.892	28.348	23.947	30.218	31.983	**32.098**
	MSSIM	0.1468	0.7852	0.6937	0.8327	0.8910	**0.9143**
	MFSIM	0.5326	0.9156	0.8878	0.9319	0.9483	**0.9561**
	SAM	41.225	8.9158	11.786	7.7761	6.3496	**5.5627**
	Time	-	30.237	34.630	970.26	5.3745	**4.5971**
Case 6	PSNR	16.239	28.755	31.594	30.830	32.212	**32.843**
	MSSIM	0.2529	0.7969	0.9136	0.8925	0.9201	**0.9242**
	MFSIM	0.6014	0.9175	0.9590	0.9228	0.9644	**0.9651**
	SAM	35.805	8.7441	5.4931	7.3799	5.4581	**5.2077**
	Time	-	29.833	34.250	969.71	5.7278	**5.3020**
Case 7	PSNR	12.891	27.691	23.759	28.871	30.103	**30.933**
	MSSIM	0.1465	0.7654	0.6623	0.7899	0.8839	**0.9066**
	MFSIM	0.5321	0.9069	0.8744	0.9142	0.9453	**0.9553**
	SAM	41.441	9.4280	11.9194	8.9173	8.1103	**7.2942**
	Time	-	30.293	34.020	972.06	5.8311	**5.1261**

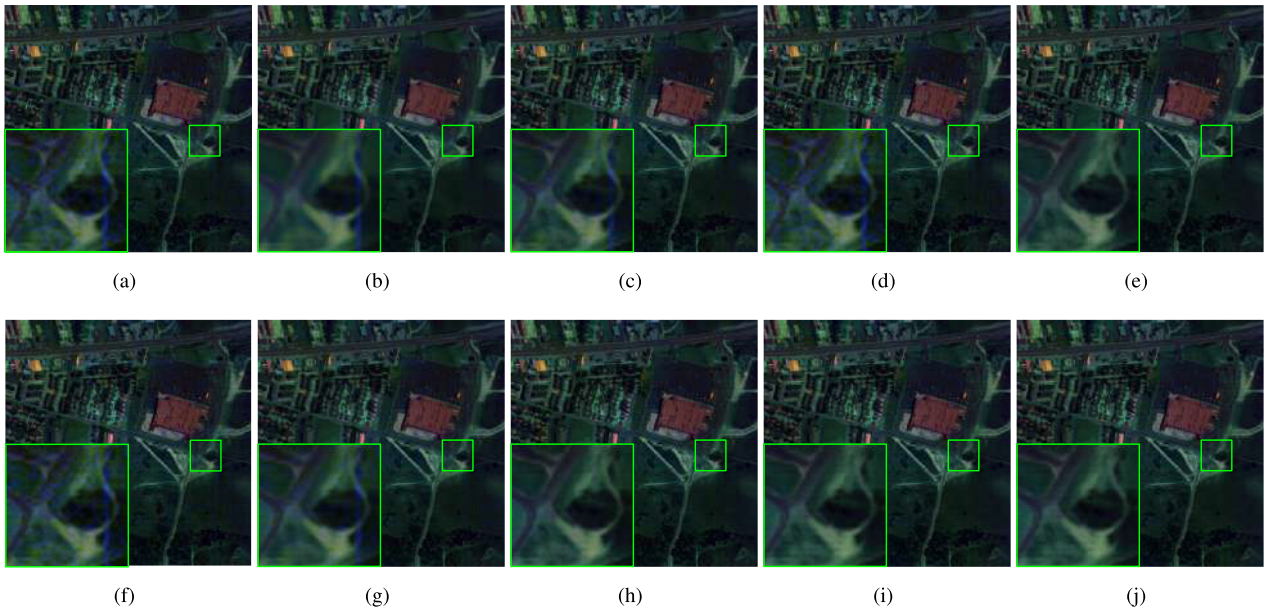


Fig. 32. Denoising for the Urban dataset. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 30, G: 110, and B: 207), (b) BM3D, (c) BM4D, (d) LRTA, (e) NAIRLMA (f), TDL, (g) KBR, (h) HyRes, (i) FastHyDe, and (j) SSSLRR-DWT.

contaminated by various types of noise, such as the Gaussian, Poisson, missing pixels, salt and pepper, and stripped noises.

2) Results on Indian Pines Dataset: Fig. 31 shows the pseudocolor images for both original and denoised Indian Pines datasets. As shown in Fig. 31, BM3D, LRTA, TDL, and KBR can remove some noise, and BM4D smooths some details. NAIRLMA, HyRes, FastHyDe, and SSSLRR-DWT achieve satisfactory denoising results. The area marked by a green box reveals that BM3D, BM4D, LRTA, TDL, and KBR have edge distortion or blurring, and NAIRLMA, HyRes, FastHyDe, and SSSLRR-DWT preserve details such as sharp and clear edges.

3) Urban Dataset: Pseudocolor images for both original corrupted by noise and strips and the denoised Urban dataset are shown in Fig. 32 for each method. The denoised images of BM3D, BM4D, LRTA, TDL, and KBR still have blue stripes. NAIRLMA, HyRes, FastHyDe, and SSSLRR-DWT remove Gaussian noise and blue strip and provide good denoised results. This also demonstrates that the proposed SSSLRR-DWT method can remove the Gaussian noise and strips in real HSI. To further illustrate the denoising effect, Fig. 33 shows the horizontal mean profiles of band 206 before and after denoising. The horizontal mean profile of a band is obtained by calculating the mean of each row. Fig. 33(a) has

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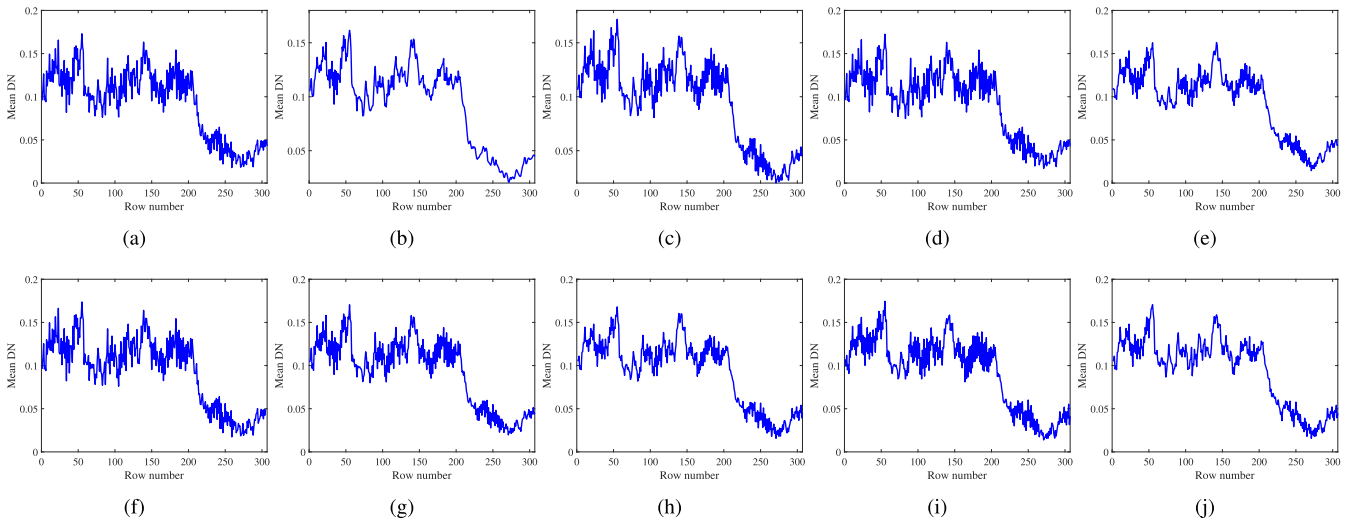


Fig. 33. Horizontal mean profiles of band 206 of the Urban dataset. (a) Original data, denoising results of (b) BM3D, (c) BM4D, (d) LRTA, (e) NAIRLMA (f), TDL, (g) KBR, (h) HyRes, (i) FastHyDe, and (j) SSSLRR-DWT.

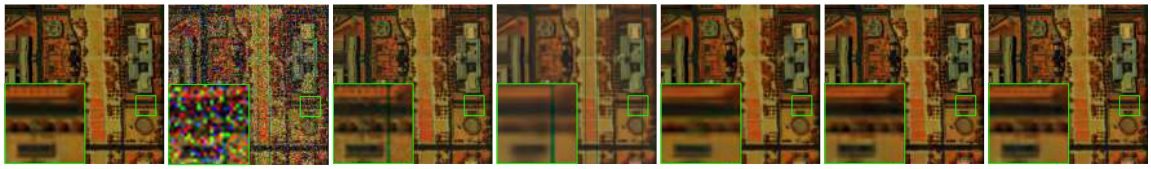


Fig. 34. Denoising for the WDCM dataset, Case 7 using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 70, G: 120, and B: 170), (b) noisy image, (c) LRMR, (d) NGmeet, (e) TLR-ℓ_0TV, (f) SSSLRR-DCT, and (g) SSSLRR-DWT.



Fig. 35. Denoising for the UP dataset, Case 7 using different methods. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 64, G: 45, and B: 10), (b) noisy image, (c) LRMR, (d) NGmeet, (e) TLR-ℓ_0TV, (f) SSSLRR-DCT, and (g) SSSLRR-DWT.

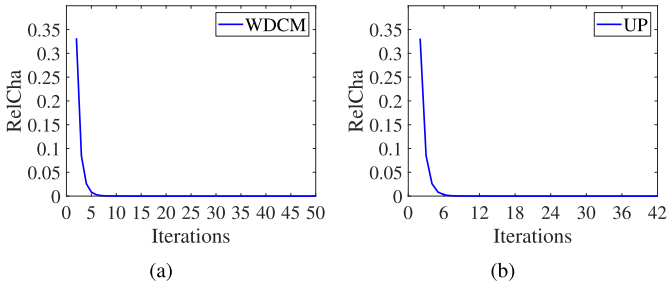


Fig. 36. RelCha values with respect to the number of iterations. (a) WDCM dataset. (b) UP dataset.

rapid fluctuations that are due to the mixed noise, and all the denoising methods diminish them. BM3D, NAIRLMA, HyRes, and SSSLRR-DWT achieve smoother curves.

C. Discussion

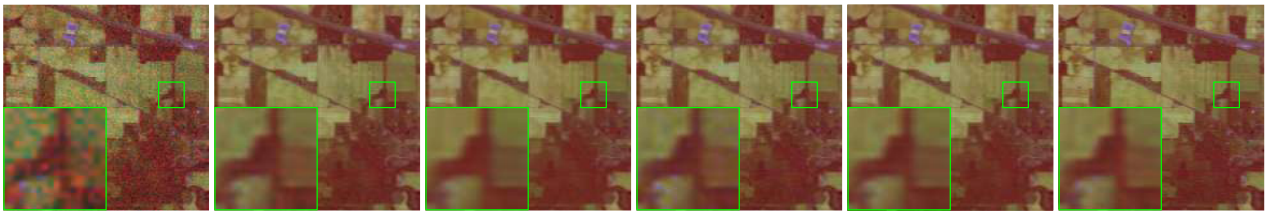
1) Convergence of the Proposed Method: The convergence is studied on the WDCM and UP datasets. Fig. 36 shows the RelCha of two denoised results for the WDCM and UP datasets for Case 2 with $\sigma = 1$ and $\eta = 10$. RelCha is the relative change between iterations, i.e., $\|\mathbf{W}^{k+1} - \mathbf{W}^k\|_F^2 / \|\mathbf{W}^k\|_F^2$, where $\mathbf{W}^k$ are the wavelet coefficient at the kth iteration. The RelCha values for SSSLRR-DWT monotonically decrease with iterations until stopping criterion is satisfied (RelCha $\leq \epsilon$).

2) Parameter Analysis: For SSSLRR-DWT, three parameters have to be obtained or chosen for the HSI denoising. Here, we analyze parameter selection for the simulated WDCM and UP datasets in Case 2 with $\sigma = 1$ and $\eta = 10$.

λ_1 is related to the strength of sparse wavelet coefficients. SSSLRR-DWT is tested using different values of λ_1 over [0.01, 0.1] and shown in Fig. 27(a). SSSLRR-DWT achieves the best performance when λ_1 is set as 0.07.

λ_ω is a constant, is related to the weight ω_i, and is correlated with the low-rank characteristic of HSI. Fig. 27(b) shows the denoised results based on the different λ_ω. SSSLRR-DWT achieves the highest PSNR when λ_ω is 0.3.

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(a) (b) (c) (d) (e) (f)

Fig. 37. Denoising for the Indian Pines dataset. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 220, G: 144, and B: 3), (b) LRMR, (c) NGmeet, (d) TLR-ℓ_0TV, (e) SSSLRR-DCT, and (f) SSSLRR-DWT.



(a) (b) (c) (d) (e) (f)

Fig. 38. Denoising for the Urban dataset. The small green square is a zoomed-in area shown in the big square. (a) Original pseudocolor image (R: 30, G: 110, and B: 207), (b) LRMR, (c) NGmeet, (d) TLR-ℓ_0TV, (e) SSSLRR-DCT, and (f) SSSLRR-DWT.

Parameter μ is used for the ADMM multipliers. Fig. 27(c) shows the robustness of the proposed SSSLRR-DWT method to the changes in μ. When μ lies in the range of [0.5, 0.6], SSSLRR-DWT achieves a high PSNR value.

3) Further Discussion: In this section, we further compare SSSLRR-DWT with four denoising methods for removing Gaussian noise with high noise level and mixed noise. The compared denoising methods are (LRMR[13]) [33], nonlocal low-rank tensor approximation (NGmeet[14]) [61], tensor low-rank constraint and ℓ_0 total variation (TLR-ℓ_0TV[15]) [62], and SSSLRR using the 3-D discrete orthogonal cosine transform (SSSLRR-DCT) instead of using the 3-D discrete orthogonal wavelet transform. The noisy simulated datasets are created as follows.

1) *Case 4:* A zero-mean Gaussian noise is added to each band with standard deviation σ drawn from a uniform distribution between 0.1 and 0.4 [$\sigma \sim U(0.1, 0.4)$].
2) *Case 5 (Gaussian Noise + Salt and Pepper Noise):* The Gaussian noise is added in the same way as Case 3. The salt and pepper noise is added to all the bands with noise density 10%.
3) *Case 6 (Gaussian Noise + Stripe):* The Gaussian noise is added in the same way as Case 3; 30% of the bands are randomly selected to add the stripes. The number of stripes for the selected band is randomly sampled from 1 to 10.
4) *Case 7 (Gaussian Noise + Salt and Pepper Noise + Stripe):* The Gaussian noise and salt and pepper noise are added in the same way as Case 5. The stripes are added in the same way as in Case 6.

[13]https://sites.google.com/site/rshewei/home
[14]https://sites.google.com/site/rshewei/home
[15]https://openremotesensing.net/kb/codes/denoising/

The simulated WDCM and UP datasets for Cases 1–7 and the real Indian Pines and Urban datasets are used in the experiments. Tables V–VIII present the PSNR, MSSIM, MFSIM, SAM, and running time of SSSLRR-DWT and all comparison methods for the WDCM and UP datasets. The highest values of PSNR, MSSIM, and MFSIM and the lowest values of SAM and running time are represented in bold. It can be seen that SSSLRR-DWT yields the best denoising results in most cases. NGmeet yields good denoising results for low Gaussian noise cases. SSSLRR-DWT is slightly better than SSSLRR-DCT. Both of them obtain good denoising results. This means that SSSLRR using both 3-D discrete orthogonal wavelet transform and 3-D discrete orthogonal cosine transform can achieve good denoising performance. Figs. 34, 35, 37, and 38 show the pseudocolor images of the simulated WDCM and UP datasets for Case 7 and the real Indian Pines and Urban datasets denoised by five different methods. The same area of each subfigure marked with a green box was enlarged for a better comparison. From Figs. 34 and 35, we can see that LRMR can remove most stripes, but there is still Gaussian noise and salt and pepper noise in the denoised image. NGmeet removes the Gaussian noise and salt and pepper noise effectively for the WDCM dataset and most Gaussian noise and salt and pepper noise for the UP dataset and cannot remove stripes and gives an over smoothed denoised image. TLR-ℓ_0TV, SSSLRR-DCT, and SSSLRR-DWT can remove the Gaussian noise, salt and pepper noise, and stripe efficiently and give good denoising results. From Figs. 37 and 38, we can see that all the five denoising methods achieve satisfactory denoising results for the real Indian Pines and Urban datasets.

4) Running Time: The running times (in seconds) for SSSLRR-DWT and the competitive methods applied to the WDCM and the UP datasets are evaluated in this section. All the experiments are run on a Linux computer using

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MATLAB R2019b. The computer has an eight-core Intel CPU 3.2 GHz and 64-GB RAM. Tables III–VIII show the running times. FastHyDe is the fastest method using less than 1 s for denoising the WDCM and the UP datasets, while KBR is the slowest method. SSSLRR-DWT has an average running time in the middle between the slow group methods (BM4D and KBR) and the fast group methods (FastHyDe, HyRes, and LRTA) and is 24.97 s for the WDCM dataset and 7.24 s for the UP dataset. The computational complexity of our method is as follows. The ADMM is an iterative algorithm. For an HSI matrix of size $m \times n \times B$, the computational complexity at each iteration is dominated by the SVD operation, which needs $O(m \times n \times B^2)$ flops. Thus, $T \times O(m \times n \times B^2)$ flops are needed for HSI, where T is the number of the iteration steps.

IV. Conclusion

In this article, a denoising method based on SSSLRR using the 3-D orthogonal transform (3-DOT) was proposed. SSSLRR can be effectively used to remove the Gaussian and mixed noise. 3-DOT is used for SSSLRR to decompose noisy HSI to sparse transform coefficients. The 3-D discrete orthogonal wavelet transform (3-D DWT) and 3-D discrete orthogonal cosine transform (3-D DCT) are two examples of 3-DOT appropriate for denoising since they concentrate the signal in few transform coefficients. SSSLRR-DWT uses the 3-D DWT to be the 3-DOT. SSSLRR-DWT transforms the original HSI to the wavelet domain. A weighted nuclear norm low-rank regularizer was proposed for the global image, which gives more flexibility than the nuclear norm. The weighted nuclear norm low-rank regularizer uses different weights to shrink the singular values of the wavelet coefficient matrix.

HSI has sparse and low-rank characteristics and can be preserved using the ℓ_1 penalty and the weighted nuclear norm low-rank penalty, respectively. Thus, a novel sparse and low-rank penalized model was proposed using both the ℓ_1 penalty and the weighted nuclear norm low-rank penalty. SSSLRR-DWT uses the sparse and low-rank penalized model can adaptively shrink wavelet coefficients and penalize the singular values of the wavelet coefficient matrix. Thus, SSSLRR-DWT uses the sparse and low-rank penalized model to remove noise effectively and recover the denoised image. The algorithm for SSSLRR-DWT was developed using ADMM. To analyze the effectiveness of different penalties for SSSLRR-DWT both in signal and wavelet domain, four different methods were analyzed. The analysis shows that SSSLRR-DWT that uses both the ℓ_1 penalty and weighted nuclear norm low-rank penalty in the wavelet domain yields better denoising results than the other methods. Moreover, a new SSSLRR-DCT denoising method was also proposed, which uses the 3-D discrete orthogonal cosine transform (3-D DCT) for SSSLRR. The analysis shows that both SSSLRR-DWT and SSSLRR-DCT have good denoising results. The experimental results indicate that our proposed method improves the denoising performance for both simulated and real noisy HSI datasets.

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